

Unmanned aerial vehicle

An **unmanned aerial vehicle** (UAV), or **unmanned aircraft system** (UAS), commonly known as an **aerial drone** or simply **drone**, is an aircraft with no human pilot, crew, or passengers on board, which instead is either autonomous or controlled remotely.^{[1][2]} UAVs were originally developed through the twentieth century for military missions too "dull, dirty or dangerous"^[3] for humans, and by the twenty-first had become essential assets to most militaries. As control technologies improved and costs fell, their use expanded to many non-military applications.^[4] These include aerial photography, area coverage,^[5] precision agriculture, forest fire monitoring,^[6] river monitoring,^{[7][8]} environmental monitoring,^{[9][10][11][12]} weather observation, policing and surveillance, infrastructure inspections, smuggling,^[13] product deliveries, entertainment, and drone racing.

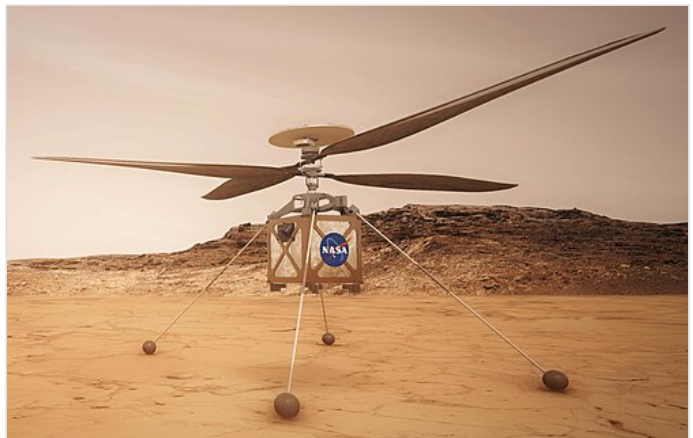
Terminology

Many terms are used for aircraft which fly without any persons on board.

Unmanned aerial vehicle (UAV) is defined as a "powered, aerial vehicle that does not carry a human operator, uses aerodynamic forces to provide vehicle lift, can fly autonomously or be piloted remotely, can be expendable or recoverable, and can carry a lethal or nonlethal payload".^[14] UAV is a term that is commonly applied in military use cases.^[15] A missile with a warhead is generally not considered a UAV because the vehicle itself is a munition, but certain types of propeller-based missile are often called "kamikaze drones" by the public and media. Also, the relation of UAVs to remote controlled model aircraft is unclear in some jurisdictions. The US FAA now defines any unmanned flying craft as a UAV regardless of mass.^[16] Similar terms are ***remotely piloted aircraft (RPA)*** and ***remotely piloted aerial vehicle (RPAV)***.



A General Atomics MQ-9 Reaper, an unmanned combat aerial vehicle used for both intelligence, surveillance, target acquisition, and reconnaissance (ISTAR) and drone strike missions.



Ingenuity is an extraterrestrial autonomous UAV that performed the first powered controlled flight by an aircraft on another planet.

UAVs and RPAVs can also be seen as components of **unmanned aircraft systems (UASs)**, which also include ground-based controllers and systems of communications with the aircraft.^[6] UAS was adopted by the United States Department of Defense (DoD) and the United States Federal Aviation Administration (FAA) in 2005 according to their Unmanned Aircraft System Roadmap 2005–2030.^[17] The International Civil Aviation Organization (ICAO) and the British Civil Aviation Authority adopted this term, also used in the European Union's Single European Sky (SES) Air Traffic Management (ATM) Research (SESAR Joint Undertaking) roadmap for 2020.^[18] This term emphasizes the importance of elements other than the aircraft. It includes elements such as ground control stations, data links and other support equipment. Similar terms are **unmanned aircraft vehicle system (UAVS)** and **remotely piloted aircraft system (RPAS)**.^[19] Many similar terms are in use. Under new regulations which came into effect 1 June 2019, RPAS has been adopted by the Canadian government to mean "a set of configurable elements consisting of a remotely piloted aircraft, its control station, the command and control links and any other system elements required during flight operation".^[20]

In common usage, *drone* is often applied to both military and civilian UAVs, while technical and regulatory documents may prefer terms such as UAV, UAS, RPAS, or *uncrewed aircraft*. The term *drone* has been used from the early days of aviation, sometimes being applied to remotely flown target aircraft used for practice firing of a battleship's guns, such as the 1920s Fairey Queen biplane floatplane and 1930s de Havilland Queen Bee biplane. Later examples included the Airspeed Queen Wasp and Miles Queen Martinet, before ultimate replacement by the GAF Jindivik.^[21] The term remains in common use. The term *uncrewed* is sometimes used rather than *unmanned* when referring to UAVs.^{[22][23][24]}



DJI Phantom, a small civilian quadcopter designed for commercial and recreational aerial photography. This type of craft is also used or adapted for surveillance by law enforcement and military organisations.



DJI Mini, a consumer-grade drone used for recreational flying

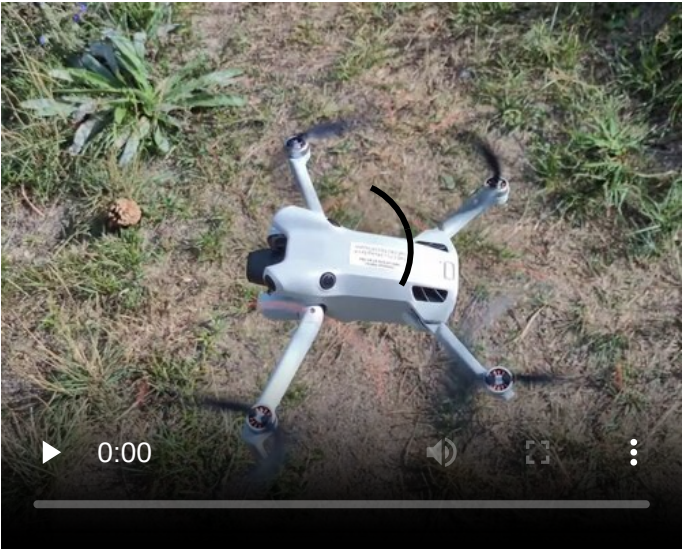


Although most large military UAVs are fixed-wing aircraft, rotorcraft designs (i.e., RUAVs) such as this Schiebel Camcopter S-100 are also used.

Autonomous drones also employ various advanced technologies to carry out their missions without human intervention, such as cloud computing, computer vision, artificial intelligence, machine learning, deep learning, and thermal sensors.^[25] For recreational uses, an aerial photography drone has first-person video, autonomous capabilities, or both.^[26]

Classification types

UAVs may be classified like any other aircraft, according to design configuration such as weight or engine type, maximum flight altitude, degree of operational autonomy, operational role, etc. According to the United States Department of Defense, UAVs are classified into five categories below:^{[27][28]}



Hovering drone in windy weather.

Group:	Group 1	Group 2	Group 3	Group 4	Group 5
Size	Small	Medium	Large	Larger	Largest
Max takeoff weight	< 20 lb (9.1 kg)	> 20 & < 55 (25 kg)	> 55 & < 1320 (600 kg)	> 1,320 lb (600 kg)	> 1,320 lb (600 kg)
Operating altitude	< 1,200 ft (370 m)	< 3,500 ft (1,100 m)	< 18,000 ft (5,500 m)	< 18,000 ft (5,500 m)	> 18,000 ft (5,500 m)
Speed	< 100 kn (190 km/h)	< 250 kn (460 km/h)	< 250 kn (460 km/h)	Any speed	Any speed

Other classifications of UAVs include:^[27]

Range and endurance

There are usually five categories when UAVs are classified by range and endurance:^[27]

Range category	Very close	Close	Short	Medium	Long
Range (km):	< 5	> 5 & < 50	> 50 & < 150	> 150 & < 650	> 650
Endurance (hr):	0.5–0.75	1–6	8–12	12–36 or 48	> 36 or 48

Size

There are usually four categories when UAVs are classified by size, with at least one of the dimensions (length or wingspan) meet the following respective limits:^[27]

Category	Micro/Very small	Mini/Small	Medium	Large
Length/wingspan:	< 50 cm	> 50 cm & < 2 m	5–10 m	> 10 m

Weight

Based on their weight, drones can be classified into five categories:

Drone categories^[29]

Category	Nano	Micro (MAV)	Miniature or Small (SUAV)	Medium	Large
Weight:	< 250 g	≥ 250 g & < 2 kg	≥ 2 kg & < 25 kg	≥ 25 kg & < 150 kg	≥ 150 kg

NATO uses a similar classification shown below:^[1]

Degree of autonomy

Drones can also be classified based on the degree of autonomy in their flight operations. ICAO classifies unmanned aircraft as either remotely piloted aircraft or fully autonomous.^[30] Some UAVs offer intermediate degrees of autonomy. For example, a vehicle may be remotely piloted in most contexts but have an autonomous return-to-base operation. Some aircraft types may optionally fly manned or as UAVs, which may include manned aircraft transformed into manned or optionally piloted UAVs (OPVs). The flight of UAVs may operate under remote control by a human operator, as **remotely piloted aircraft (RPA)**, or with various degrees of autonomy, such as autopilot assistance, up to fully autonomous aircraft that have no provision for human intervention.^{[31][32]}

NATO UAS Classification						
Class	Category	Normal Employment	Normal Operating Altitude	Normal Mission Radius	Primary Supported Commander	Example Platform
Class III (≥ 600 kg)	Strike/ Combat	Strategic/ National	Up to 65,000 ft MSL	Unlimited (BLOS)	Theatre	Reaper
	HALE	Strategic/ National	Up to 65,000 ft MSL	Unlimited (BLOS)	Theatre	Global Hawk
	MALE	Operational/ Theatre	Up to 45,000 ft MSL	Unlimited (BLOS)	JTF	Heron
Class II (150 kg–600 kg)	Tactical	Tactical Formation	Up to 18,000 ft AGL	200 km (LOS)	Division, Brigade	Watchkeeper
Class I (≤ 150 kg)	Small (≥ 15 kg)	Tactical Unit	Up to 5,000 ft AGL	50 km (LOS)	Battalion, Regiment	Scan Eagle
	Mini (≤ 15 kg)	Tactical Sub-unit (manual or hand launch)	Up to 3,000 ft AGL	Up to 25 km (LOS)	Company, Platoon, Squad	Skylark
	Micro (≤ 66 lb)	Tactical Sub-unit (manual or hand launch)	Up to 200 ft AGL	Up to 5 km (LOS)	Platoon, Squad	Black Widow

NATO classification of unmanned aerial vehicles

Altitude

Based on the altitude, the following UAV classifications have been used at industry events such as ParcAberporth Unmanned Systems forum:

- Hand-held 2,000 ft (600 m) altitude, about 2 km range
- Close 5,000 ft (1,500 m) altitude, up to 10 km range
- NATO type 10,000 ft (3,000 m) altitude, up to 50 km range
- Tactical 18,000 ft (5,500 m) altitude, about 160 km range
- MALE (medium altitude, long endurance) up to 30,000 ft (9,000 m) and range over 200 km
- HALE (high altitude, long endurance) over 30,000 ft (9,100 m) and indefinite range
- Hypersonic high-speed, supersonic (Mach 1–5) or hypersonic (Mach 5+) 50,000 ft (15,200 m) or suborbital altitude, range over 200 km
- Orbital low Earth orbit (Mach 25+)
- CIS lunar Earth-Moon transfer
- Computer-assisted carrier guidance system (CACGS) for UAVs

Composite criteria

An example of classification based on the composite criteria is U.S. Military's unmanned aerial systems (UAS) classification of UAVs based on weight, maximum altitude and speed of the UAV component.

Lift type

FFLO

Forward flight lift only classified by ICAO with the aircraft type designator FFLO - UAV refers to plane-like drones

VFHC

Vertical flight/hover capability classified by ICAO with the aircraft type designator VFHC - UAV refers to drones using rotating blades for lift as well as motion

Power sources

UAVs can be classified based on their power or energy source, which significantly impacts their flight duration, range, and environmental impact. The main categories include:

- **Battery-powered (electric):** These UAVs use rechargeable batteries, offering quiet operation and lower maintenance but potentially limited flight times. The reduced noise levels make them suitable for urban environments and sensitive operations.^[33]
- **Fuel-powered (internal combustion):** Utilizing traditional fuels like gasoline or diesel, these UAVs often have longer flight times but may be noisier and require more maintenance. They are typically used for applications requiring extended endurance or heavy payload capacity.^[34]
- **Hybrid:** Combining electric and fuel power sources, hybrid UAVs aim to balance the benefits of both systems for improved performance and efficiency. This configuration could allow for versatility in mission profiles and adaptability to different operational requirements.^[35]
- **Hydrogen fuel cell:** hydrogen fuel cells offer the potential for longer flight times than batteries yet stealthier (no heat signature) operation than combustion engines.^[36] The high energy density of hydrogen makes it a promising option for future UAV propulsion systems.^[37]
- **Solar-powered:** Equipped with solar panels, these UAVs can potentially achieve extended flight times by harnessing solar energy, especially at high altitudes. Solar-powered UAVs may be particularly suited for long-endurance missions and environmental monitoring applications.^[38]
- **Nuclear-powered:** While nuclear power has been explored for larger aircraft, its application in UAVs remains largely theoretical due to safety concerns and regulatory challenges. Research in this area is ongoing but faces significant hurdles before practical implementation.^[39]

History

1800's

Early drones

The earliest recorded use of an unmanned aerial vehicle for warfighting occurred in July 1849,^[41] with a balloon carrier (the precursor to the aircraft carrier)^[42] in the first offensive use of air power in naval aviation.^{[43][44][45]} Austrian forces besieging Venice attempted to launch some 200 incendiary balloons at the besieged city. The balloons were launched mainly from land; however, some were also launched from the Austrian ship SMS Vulcano. At least one bomb fell in the city; however, due to the wind changing after launch, most of the balloons missed their target, and some drifted back over Austrian lines and the launching ship Vulcano.^{[46][47][48]}

The Spanish engineer Leonardo Torres Quevedo introduced a radio-based control-system called the Telekino^[49] at the Paris Academy of Science in 1903, as a way of testing airships without risking human life.^{[50][51][52]}

1900s

Significant development of drones started in the 1900s, and originally focused on providing practice targets for training military personnel. The earliest attempt at a powered UAV was A. M. Low's "Aerial Target" in 1916.^[53] Low confirmed that Geoffrey de Havilland's monoplane was the one that flew under control on 21 March 1917 using his radio system.^[54] Following this successful demonstration in the spring of 1917 Low was transferred to develop aircraft controlled fast motor launches D.C.B.s with the Royal Navy in 1918 intended to attack shipping and port installations and he also assisted Wing Commander Brock in preparations for the Zeebrugge Raid. Other British unmanned developments followed, leading to the fleet of over 400 de Havilland 82 Queen Bee aerial targets that went into service in 1935.

Nikola Tesla described a fleet of uncrewed aerial combat vehicles in 1915.^[55] These developments also inspired the construction of the Kettering Bug by Charles Kettering from Dayton, Ohio and the Hewitt-Sperry Automatic Airplane – initially meant as an uncrewed plane that would carry an explosive payload to a predetermined target. Development continued during World War I, when the Dayton-Wright Airplane Company invented a pilotless aerial torpedo that would explode at a preset time.^[56]



Winston Churchill and others waiting to watch the launch of a de Havilland Queen Bee target drone, 6 June 1941



A Ryan Firebee, one of a series of target drones/unpiloted aerial vehicles that first flew in 1951. Israeli Air Force Museum, Hatzerim airbase, Israel, 2006



Last preparations before the first tactical UAV mission across the Suez Canal (1969). Standing: Major Shabtai Brill from the Israeli Intelligence Corps, the innovator of the tactical UAV.

The film star and model-airplane enthusiast Reginald Denny developed the first scaled remote piloted vehicle in 1935.^[53]

Soviet researchers experimented with controlling Tupolev TB-1 bombers remotely in the late 1930s.^[57]

World War II

In 1940, Reginald Denny started the Radioplane Company and more models emerged during World War II – used both to train antiaircraft gunners and to fly attack-missions. Nazi Germany produced and used various UAV aircraft during the war, like the Argus As 292 and the V-1 flying bomb with a jet engine. Fascist Italy developed a specialised drone version of the Savoia-Marchetti SM.79 flown by remote control, although the Armistice with Italy was enacted prior to any operational deployment.^[58]



The Israeli Tadiran Mastiff, which first flew in 1975, is seen by many as the first modern battlefield UAV, due to its data-link system, endurance-loitering, and live video-streaming.^[40]

Postwar period

After World War II development continued in vehicles such as the American JB-4 (using television/radio-command guidance), the Australian GAF Jindivik and Teledyne Ryan Firebee I of 1951, while companies like Beechcraft offered their Model 1001 for the U.S. Navy in 1955.^[53] Nevertheless, they were little more than remote-controlled airplanes until the Vietnam War. In 1959, the U.S. Air Force, concerned about losing pilots over hostile territory, began planning for the use of uncrewed aircraft.^[59] Planning intensified after the Soviet Union shot down a U-2 in 1960. Within days, a highly classified UAV program started under the code name of "Red Wagon".^[60] The August 1964 clash in the Tonkin Gulf between naval units of the U.S. and the North Vietnamese Navy initiated America's highly classified UAVs (Ryan Model 147, Ryan AQM-91 Firefly, Lockheed D-21) into their first combat missions of the Vietnam War.^[61] When the Chinese government^[62] showed photographs of downed U.S. UAVs via Wide World Photos,^[63] the official U.S. response was "no comment".

During the War of Attrition (1967–1970) in the Middle East, Israeli intelligence tested the first tactical UAVs installed with reconnaissance cameras, which successfully returned photos from across the Suez Canal. This was the first time that tactical UAVs that could be launched and landed on any short runway (unlike the heavier jet-based UAVs) were developed and tested in battle.^[64]

In the 1973 Yom Kippur War, Israel used UAVs as decoys to spur opposing forces into wasting expensive anti-aircraft missiles.^[65] After the 1973 Yom Kippur War, a few key people from the team that developed this early UAV joined a small startup company that aimed to develop UAVs into a commercial product, eventually purchased by Tadiran and leading to the development of the first Israeli UAV.^[66]

In 1973, the U.S. military officially confirmed that they had been using UAVs in Southeast Asia (Vietnam).^[67] Over 5,000 U.S. airmen had been killed and over 1,000 more were missing or captured. The USAF 100th Strategic Reconnaissance Wing flew about 3,435 UAV missions during the war^[68] at a cost of about 554 UAVs lost to all causes. In the words of USAF General George S. Brown, Commander, Air Force Systems Command, in 1972, "The only reason we need (UAVs) is that we don't want to

needlessly expend the man in the cockpit."^[69] Later that year, General John C. Meyer, Commander in Chief, Strategic Air Command, stated, "we let the drone do the high-risk flying ... the loss rate is high, but we are willing to risk more of them ...they save lives!"^[69]

During the 1973 Yom Kippur War, Soviet-supplied surface-to-air missile-batteries in Egypt and Syria caused heavy damage to Israeli fighter jets. As a result, Israel developed the IAI Scout as the first UAV with real-time surveillance.^{[70][71][72]} The images and radar decoys provided by these UAVs helped Israel to completely neutralize the Syrian air defenses at the start of the 1982 Lebanon War, resulting in no pilots downed.^[73] In Israel in 1987, UAVs were first used as proof-of-concept of super-agility, post-stall controlled flight in combat-flight simulations that involved tailless, stealth-technology-based, three-dimensional thrust vectoring flight-control, and jet-steering.^[74]

Post-Cold War UAVs

With the maturing and miniaturization of applicable technologies in the 1980s and 1990s, interest in UAVs grew within the higher echelons of the U.S. military. The U.S. funded the Counterterrorism Center (CTC) within the CIA, which sought to fight terrorism with the aid of modernized drone technology.^[75] In the 1990s, the U.S. DoD gave a contract to AAI Corporation along with Israeli company Malat. The U.S. Navy bought the AAI Pioneer UAV that AAI and Malat developed jointly. Many of these UAVs saw service in the 1991 Gulf War. UAVs demonstrated the possibility of cheaper, more capable fighting-machines, deployable without risk to aircrews. Initial generations primarily involved surveillance aircraft, but some carried armaments, such as the General Atomics MQ-1 Predator, that launched AGM-114 Hellfire air-to-ground missiles.



The Turkish STM Kargu was the first  lethal autonomous weapon to attack enemy combatants in warfare.

2000s

CAPECON, a European Union project to develop UAVs,^[76] ran from 1 May 2002 to 31 December 2005.^[77]

As of 2012, the United States Air Force (USAF) employed 7,494 UAVs – almost one in three USAF aircraft.^{[78][79]} The Central Intelligence Agency also operated UAVs.^[80] By 2013 at least 50 countries used UAVs. China, Iran, Israel, Pakistan, Turkey, and others designed and built their own varieties. The use of drones has continued to increase.^[81] Due to their wide proliferation, no comprehensive list of UAV systems exists.^{[79][82]}

In 2006, the United States Federal Aviation Administration (FAA) allowed the usage of unmanned aerial vehicles inside civilian airspace with specific regulations, laying out legal groundwork for consumer drone usage inside the United States.^[83]

In 2013, DJI released the first model of the Phantom drone, DJI's fully assembled drone model. Priced at \$629, Phantom was an entry-level drone, featuring much more user-friendly experiences than other drones on the market at the time. The DJI Phantom was considered one of the most influential consumer drone products ever made. With its affordability, accessibility, and user-friendly software, it quickly

captured the consumer drone market from hobbyists, professionals, and introduced the modern aerial photography drone form factor to the general public.^{[84][85]} By 2017, DJI alone had over 75% of the global consumer drone market share.^[86]

In 2020, a Kargu 2 drone hunted down and attacked a human target in Libya, according to a report from the UN Security Council's Panel of Experts on Libya, published in March 2021. This may have been the first time an autonomous weapons system capable of lethal engagement attacked human beings.^[87]

Drone technology, including systems such as the Turkish Bayraktar TB2, has been identified as a contributing factor in Azerbaijan's military performance in the 2020 Nagorno-Karabakh war against Armenia.^[88]

NASA has sent drones to distant planets. The Ingenuity helicopter is an autonomous UAV that operated on Mars from 2021 to 2024. As of 2024 the Dragonfly spacecraft is being developed and is aiming to reach and examine Saturn's moon Titan. Its primary goal is to roam around the surface, expanding the amount of area to be researched previously seen by landers. As a UAV, Dragonfly allows examination of potentially diverse types of soil. The drone is set to launch in 2027, and is estimated to take seven more years to reach the Saturnian system.^[89]



Artist's concept of Ingenuity landing on Mars

Miniaturization is also supporting the development of small UAVs, which can be used as an individual system or in a fleet, enabling efficient surveying of large areas in a relatively small amount of time.^[90]

The April 2024 Iranian strikes against Israel took place on 13 April 2024 when the Iranian Revolutionary Guard and other groups of the Axis of Resistance launched about 300 drones at Israel, a distance of about 1,500 kilometres.^{[91][92][93][94][95]}

According to data from GlobalData, the global military uncrewed aerial systems (UAS) market, which forms a significant part of the UAV industry, is projected to experience a compound annual growth rate of 4.8% over the next decade. This represents a near doubling in market size, from \$12.5 billion in 2024 to an estimated \$20 billion by 2034.^[96]

Russo-Ukrainian war

The Russian invasion of Ukraine in 2022 was widely described as the first full-scale war featuring the large-scale use of commercial and consumer-grade UAVs in military settings, particularly quadcopters and first-person view (FPV) drones, modified with sensors and explosives for various missions and tactical usage. The consumer and small UAVs have significantly influenced modern warfare, particularly due to their accessibility and cost and contributed to the development of new offensive and defensive strategies.^{[97][98]}

Both Ukraine and Russia used small consumer drones extensively during the war, which served as a tactical surveillance, strike, and propaganda tools.^[99] Consumer drones were sourced by governments, hobbyists, international donations to Ukraine and Russia to support each side on the battlefield, and were

often flown by drone hobbyists recruited by the armed forces.^[100] Consumer drones' affordability, technical capabilities, and reliability contributed to their widespread adoption.^{[101][100]} They were also preferred for their commercial availability.^[102] Companies attempted to restrict military usage of consumer product, but effect was limited,^{[103][104][105]} as donors and buyers could ship the drones across the border via intermediates and modified their software to circumnavigate the restrictions.^{[106][107]} In Ukraine, small commercial UAVs and FPV drones had become an ubiquitous and critical part of the war.^[108]



Ukrainian drone operator controlling a 
DJI Mavic drone

Design

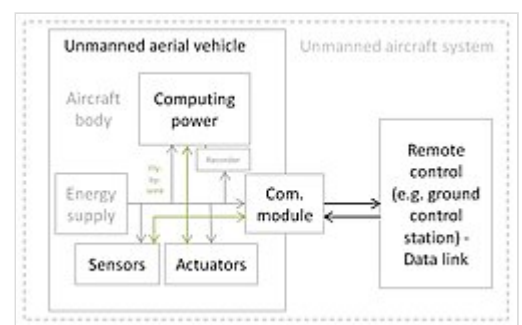
Crewed and uncrewed aircraft of the same type generally have recognizably similar physical components. The main exceptions are the cockpit and environmental control system or life-support systems. Some UAVs carry payloads (such as a camera) that weigh considerably less than an adult human, and as a result, can be considerably smaller. Though they carry heavy payloads, weaponized military UAVs are lighter than their crewed counterparts with comparable armaments.

Small civilian UAVs have no life-critical systems, and can thus be built out of lighter but less sturdy materials and shapes, and can use less robustly tested electronic control systems. For small UAVs, the quadcopter design has become popular, though this layout is rarely used for crewed aircraft. Miniaturization means that less-powerful propulsion technologies can be used that are not feasible for crewed aircraft, such as small electric motors and batteries.

Control systems for UAVs are often different from crewed craft. For remote human control, a camera and video link almost always replace the cockpit windows; radio-transmitted digital commands replace physical cockpit controls. Autopilot software is used on both crewed and uncrewed aircraft, with varying feature sets.^{[109][110][111]}

Aircraft configuration

UAVs can be designed in different configurations than manned aircraft, both because there is no need for a cockpit and its windows, and there is no need to optimize for human comfort, although some UAVs are adapted from piloted examples, or are designed for optionally piloted modes. Air safety is also less of a critical requirement for unmanned aircraft, allowing the designer greater freedom to experiment. Instead, UAVs are typically designed around their onboard payloads and their ground equipment. These factors have led to a great variety of airframe and motor configurations in UAVs.



General physical structure of a UAV 

For conventional flight, the flying wing and blended wing body offer light weight combined with low drag and stealth, and are popular configurations for many use cases. Larger types which carry a variable payload are more likely to feature a distinct fuselage with a tail for stability, control and trim, although the wing configurations in use vary widely.

For uses that require vertical flight or hovering, the tailless quadcopter requires a relatively simple control system and is common for smaller UAVs. Multirotor designs with 6 or more rotors is more common with larger UAVs, where redundancy is prioritized.^[112]

Propulsion

Traditional internal combustion and jet engines remain in use for drones requiring long range. However, for shorter-range missions electric power has almost entirely taken over. The distance record for a UAV (built from balsa wood and mylar skin) across the North Atlantic Ocean is held by a gasoline model airplane or UAV. Manard Hill "in 2003 when one of his creations flew 1,882 miles across the Atlantic Ocean on less than a gallon of fuel" holds this record.^[113]

Besides the traditional piston engine, the Wankel rotary engine is used by some drones. This type offers high power output for lower weight, with quieter and more vibration-free running. Claims have also been made for improved reliability and greater range.

Small drones mostly use lithium-polymer batteries (Li-Po)^[114], while some larger vehicles have adopted the hydrogen fuel cell.^{[115][116][117]} Hydrogen-fueled proton-exchange membrane fuel cells for UAVs have the advantages of longer flight duration than rechargeable lithium-ion batteries, of lower total cost of ownership than primary lithium metal batteries and of better stealth than heat engines.^[36]

The energy density of modern Li-Po batteries is far less than gasoline or hydrogen. However, electric motors are cheaper, lighter and quieter. Complex multi-engine, multi-propeller installations are under development with the goal of improving aerodynamic and propulsive efficiency. For such complex power installations, battery elimination circuitry (BEC) may be used to centralize power distribution and minimize heating, under the control of a microcontroller unit (MCU).

Ornithopters – wing propulsion

Flapping-wing ornithopters, imitating birds or insects, have been flown as microUAVs. Their inherent stealth recommends them for spy missions.^[118]

Sub-1g microUAVs inspired by flies, albeit using a power tether, have been able to "land" on vertical surfaces.^[119] Other projects mimic the flight of beetles and other insects.^[120]

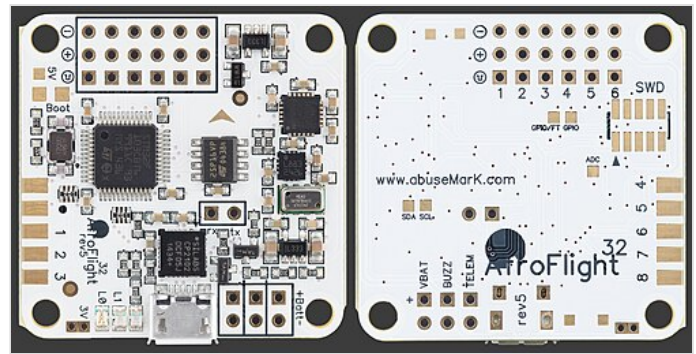


Rear view of the Iranian Shahed 149 
Gaza UCAV

Computer control systems

UAV computing capability followed the advances of computing technology, beginning with analog controls and evolving into microcontrollers, then system-on-a-chip (SOC) and single-board computers (SBC).

Modern system hardware for UAV control is often called the flight controller (FC), flight controller board (FCB) or autopilot. Common UAV-systems control hardware typically incorporate a primary microprocessor, a secondary or failsafe processor, and sensors such as accelerometers, gyroscopes, magnetometers, and barometers into a single module.



A flight controller run on either CleanFlight or BaseFlight firmware for multirotor UAVs

In 2024 EASA agreed on the first certification basis for a UAV flight controller in compliance with the ETSO-C198 for Embention's autopilot. The certification of the UAV flight control systems aims to facilitate the integration of UAVs within the airspace and the operation of drones in critical areas.^[121]

Architecture

Sensors

Position and movement sensors give information about the aircraft state. Exteroceptive sensors deal with external information like distance measurements, while exproprioceptive ones correlate internal and external states.^[122]

Non-cooperative sensors are able to detect targets autonomously so they are used for separation assurance and collision avoidance.^[123]

Degrees of freedom (DOF) refers to both the amount and quality of onboard sensors: 6 DOF implies 3-axis gyroscopes and accelerometers (a typical inertial measurement unit – IMU), 9 DOF refers to an IMU plus a compass, 10 DOF adds a barometer and 11 DOF usually adds a GPS receiver.^[124]

In addition to the navigation sensors, the UAV (or UAS) can be also equipped with monitoring devices such as: RGB, multispectral, hyper-spectral cameras or LiDAR, which may allow providing specific measurements or observations.^[125]

Actuators

UAV actuators include digital electronic speed controllers (which control the RPM of the motors) linked to motors/engines and propellers, servomotors (for planes and helicopters mostly), weapons, payload actuators, LEDs and speakers.

Software

Modern UAVs run a software stack that ranges from low-level firmware that directly controls actuators, to high level flight planning. At the lowest level, firmware directly controls reading from sensors such as an IMU^[126] and commanding actuators such as motors. Control software (often referred to as an autopilot) is responsible for computing actuator speeds given desired vehicle velocity. Due to its direct interaction with hardware, this software is time-critical and may run on microcontrollers. This software may also handle radio communications, in the case of UAVs that are not autonomous. One popular example is the PX4 autopilot.

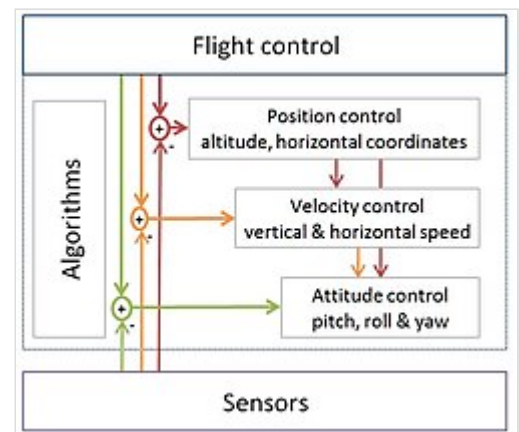
At the next level, autonomy algorithms compute the desired velocity given higher level goals. For example, trajectory optimization^[127] may be used to calculate a flight trajectory given a desired goal location. This software is not necessarily time-critical, and can often run on a single board computer running an operating system such as Linux with relaxed time constraints.

Deep reinforcement learning has also been investigated for UAV flight control, particularly for navigating continuous three-dimensional environments and improving autonomous decision-making.^[128] Similarly, AI-driven automation is increasingly being applied to counter-UAS (C-UAS) systems to reduce operator cognitive load by automatically linking electronic drone detection data with physical tracking cameras for "hands-free" target tracking.^[129]

Loop principles

UAVs employ open-loop, closed-loop or hybrid control architectures.

- Open loop – This type provides a positive control signal (faster, slower, left, right, up, down) without incorporating feedback from sensor data.
- Closed loop – This type incorporates sensor feedback to adjust behavior (reduce speed to reflect tailwind, move to altitude 300 feet). The PID controller is common. Sometimes, feedforward is employed, transferring the need to close the loop further.^[130]



Typical flight-control loops for a multirotor

Communications

UAVs use a radio for control and exchange of video and other data. Early UAVs had only narrowband uplink. Downlinks came later. These bi-directional narrowband radio links carried command and control (C&C) and telemetry data about the status of aircraft systems to the remote operator.

In most modern UAV applications, video transmission is required. So instead of having separate links for C&C, telemetry and video traffic, a broadband link is used to carry all types of data. These broadband links can leverage quality of service techniques and carry TCP/IP traffic that can be routed over the internet.

The radio signal from the operator side can be issued from either:

- Ground control – a human operating a radio transmitter/receiver, a smartphone, a tablet, a computer, or the original meaning of a military ground control station (GCS).
- Remote network system, such as satellite duplex data links for some military powers. Downstream digital video over mobile networks has also entered consumer markets, while direct UAV control uplink over the cellular mesh and LTE have been demonstrated and are in trials.^[131]
- Another aircraft, serving as a relay or mobile control station – military manned-unmanned teaming (MUM-T).^[132]

Modern networking standards have explicitly considered drones and therefore include optimizations. The 5G standard has mandated reduced user plane latency to 1ms while using ultra-reliable and low-latency communications.^[133]

UAV-to-UAV coordination supported by Remote ID communication technology. Remote ID messages (containing the UAV coordinates) are broadcast and can be used for collision-free navigation.^[134]

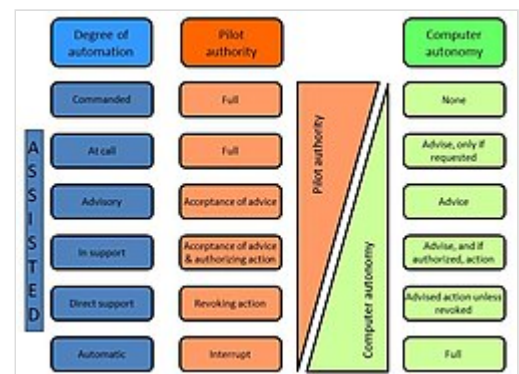
Autonomy

The level of autonomy in UAVs varies widely. UAV manufacturers often build in specific autonomous operations, such as:^[135]

- Self-level: attitude stabilization on the pitch and roll axes.
- Altitude hold: The aircraft maintains its altitude using barometric pressure and/or GPS data.
- Hover/position hold: Keep level pitch and roll, stable yaw heading and altitude while maintaining position using GNSS or inertial sensors.
- Headless mode: Pitch control relative to the position of the pilot rather than relative to the vehicle's axes.
- Care-free: automatic roll and yaw control while moving horizontally
- Takeoff and landing (using a variety of aircraft or ground-based sensors and systems; see also "autoland")
- Failsafe: automatic landing or return-to-home upon loss of control signal
- Return-to-home: Fly back to the point of takeoff (often gaining altitude first to avoid possible intervening obstructions such as trees or buildings).
- Follow-me: Maintain relative position to a moving pilot or other object using GNSS, image recognition or homing beacon.
- GPS waypoint navigation: Using GNSS to navigate to an intermediate location on a travel path.
- Orbit around an object: Similar to *follow-me* but continuously circle a target.
- Pre-programmed aerobatics (such as rolls and loops)
- Pre-programmed delivery (delivery drones)

One approach to quantifying autonomous capabilities is based on OODA terminology, as suggested by a 2002 US Air Force Research Laboratory report, and used in the table on the right.^[136]

Full autonomy is available for specific tasks, such as airborne refueling^[137] or ground-based battery switching.



UAV's degrees of autonomy

Other functions available or under development include; collective flight, real-time collision avoidance, wall following, corridor centring, simultaneous localization and mapping and swarming,^[138] cognitive radio, and machine learning. In this context, computer vision can play an important role for automatically ensuring flight safety.

Performance considerations

Flight envelope

UAVs can be programmed to perform aggressive maneuvers or landing/perching on inclined surfaces,^[139] and then to climb toward better communication spots.^[140] Some UAVs can control flight with varying flight modelisation,^{[141][142]} such as VTOL designs.

UAVs can also implement perching on a flat vertical surface.^[143]

Endurance

UAV endurance is not constrained by the physiological capabilities of a human pilot.

Because of their small size, low weight, low vibration and high power-to-weight ratio, Wankel rotary engines are used in many large UAVs. Their engine rotors cannot seize; the engine is not susceptible to shock-cooling during descent and it does not require an enriched fuel mixture for cooling at high power. These attributes reduce fuel usage, increasing range or payload.

Proper drone cooling is essential for long-term drone endurance. Overheating and subsequent engine failure is the most common cause of drone failure.^[144]

Hydrogen fuel cells, using hydrogen power, may be able to extend the endurance of small UAVs, up to several hours.^{[145][146]}

Micro air vehicles endurance is so far best achieved with flapping-wing UAVs, followed by planes and multirotors standing last, due to lower Reynolds number.^[122]

Solar-electric UAVs, a concept originally championed by the AstroFlight Sunrise in 1974, have achieved flight times of several weeks.

Solar-powered atmospheric satellites ("atmosats") designed for operating at altitudes exceeding 20 km (12 miles, or 60,000 feet) for as long as five years could potentially perform duties more economically and with more versatility than low Earth orbit satellites. Likely applications include weather drones for weather monitoring, disaster recovery, Earth imaging and communications.



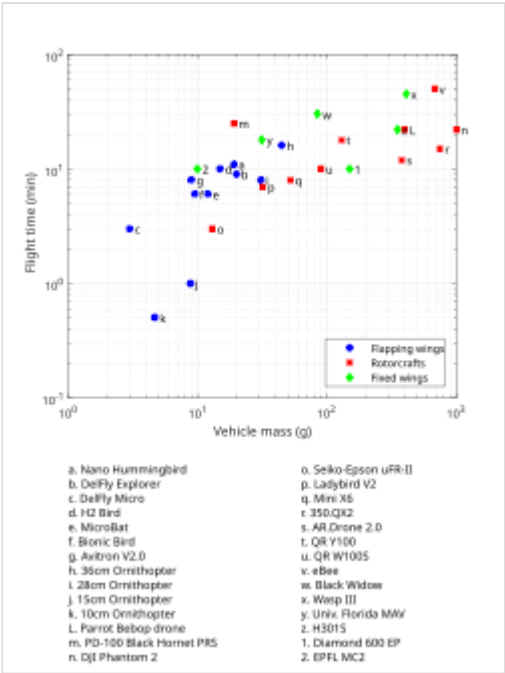
A Northrop Grumman X-47B unmanned combat aircraft demonstrator of the US Navy refuels in flight from a tanker aircraft. 



UEL UAV-741 Wankel engine for UAV operations, used on AAI RQ-7 Shadow 

Electric UAVs powered by microwave power transmission or laser power beaming are other potential endurance solutions.^[147]

Another application for a high endurance UAV would be to "stare" at a battlefield for a long interval (ARGUS-IS, Gorgon Stare, Integrated Sensor Is Structure) to record events that could then be played backwards to track battlefield activities.



Flight time against mass of small (less than 1 kg) drones^[122]

Lengthy endurance flights			
UAV	Flight time hours:minutes	Date	Notes
<u>Boeing Condor</u>	58:11	1989	The aircraft is currently in the <u>Hiller Aviation Museum</u> . [148]
<u>General Atomics Gnat</u>	40:00	1992	[149][150]
<u>TAM-5</u>	38:52	11 August 2003	Smallest UAV to cross the Atlantic [151]
<u>QinetiQ Zephyr Solar Electric</u>	54:00	September 2007	[152][153]
<u>RQ-4 Global Hawk</u>	33:06	22 March 2008	Set an endurance record for a full-scale, operational uncrewed aircraft. [154]
<u>QinetiQ Zephyr Solar Electric</u>	82:37	28–31 July 2008	[155]
<u>QinetiQ Zephyr 7</u>	336:22	9–23 July 2010	Solar electric powered. Remained aloft for 14 days. Also filed for FAI altitude record of 70,740 ft (21,561 m) [156]

The delicacy of the British PHASA-35 military drone (at a late stage of development) is such that traversing the first turbulent twelve miles of atmosphere is a hazardous endeavor. It has, however, remained on station at 65,000 feet for 24 hours. Airbus' Zephyr in 2023 has attained 70,000 feet and flown for 64 days; 200 days aimed at. This is sufficiently close enough to near-space for them to be regarded in "pseudo-satellites" as regards to their operational capabilities.^[157]

Reliability

Reliability improvements target all aspects of UAV systems, using resilience engineering and fault tolerance techniques.

Individual reliability covers robustness of flight controllers, to ensure safety without excessive redundancy to minimize cost and weight.^[158] Besides, dynamic assessment of flight envelope allows damage-resilient UAVs, using non-linear analysis with ad hoc designed loops or neural networks.^[159] UAV software liability is bending toward the design and certifications of crewed avionics software.^[160]

Swarm resilience involves maintaining operational capabilities and reconfiguring tasks given unit failures.^[161]

Applications

In recent years, autonomous drones have begun to transform various application areas as they can fly beyond visual line of sight (BVLOS)^[162] while maximizing production, reducing costs and risks, ensuring site safety, security and regulatory compliance,^[163] and protecting the human workforce in times of a pandemic.^[164] They can also be used for consumer-related missions like package delivery, as demonstrated by Amazon Prime Air, and critical deliveries of health supplies.

There are numerous civilian, commercial, military, and aerospace applications for UAVs.^[4] These include:

General

Recreation, disaster relief, archeology, conservation of biodiversity and habitat,^[165] law enforcement, crime, and terrorism.

Commercial

Aerial surveillance, filmmaking,^[166] journalism, scientific research, surveying, cargo transport, mining, manufacturing, forestry, solar farming, thermal energy, ports and agriculture.

Warfare

As of 2020, seventeen countries have armed UAVs, and more than 100 countries use UAVs in a military capacity.^[169] The first five countries producing domestic UAV designs are Turkey, United States, China, Israel and Iran.^{[170][171][172][173][174]} Top military UAV manufactures include Baykar,^{[170][175][172]} General Atomics, Elbit Systems, Rafael Advanced Defense Systems, Lockheed Martin, Northrop Grumman, Boeing, Turkish Aerospace Industries, IAIO, CASC and CAIG.^[174] China has established and expanded its presence in military UAV market^[174] since 2010. In the early 2020s, Turkey also established and expanded its presence in the military UAV market.^{[171][174][172][175]}



An Iranian Qods Mohajer-6 UCAV of the Islamic Revolutionary Guard Corps Ground Forces equipped with two Qaem precision-guided munition 

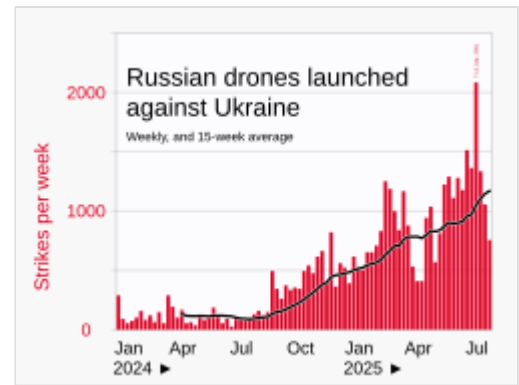
In the early 2010s, Israeli companies mainly focused on small surveillance UAV systems, and by the number of drones, Israel exported 60.7% (2014) of UAVs on the market while the United States exported 23.9% (2014).^[176] Between 2010 and 2014, there were 439 drones exchanged compared to 322 in the five years previous to that, among these only small fraction of overall trade – just 11 (2.5%) of the 439 are armed drones.^[176] The US alone operated over 9,000 military UAVs in 2014; among them more than 7000 are RQ-11 Raven miniature UAVs.^[177] Since 2010, Chinese drone companies have begun to export large quantities of drones to the global military market. Of the 18 countries that are known to have received military drones between 2010 and 2019, the top 12 all purchased their drones from China.^{[174][178]} The shift accelerated in the 2020s due to China's advancement in drone technologies and manufacturing, compounded by market demand from the Russo-Ukrainian war and the Gaza war.^{[179][180][181]}

For intelligence and reconnaissance missions, the inherent stealth of micro UAV flapping-wing ornithopters, imitating birds or insects, offers potential for covert surveillance and makes them difficult targets to bring down.

Unmanned surveillance and reconnaissance aerial vehicle are used for reconnaissance, attack, demining, and target practice.^[182]

Since the start of the Russo-Ukrainian war, a dramatic increase in UAV development took place with Ukraine creating the Brave1 platform to promote rapid development of innovative systems. By 2025, Ukraine and Russia had both established their own dedicated UAV military commands, named the "Unmanned Systems Forces" (literally "Pilotless Systems Forces"^[a]).^[183] Unlike Russia, Ukraine has been able to innovatively employ UAVs to deliver equipment like flexible medical stretchers or electric bicycles to help evacuate wounded soldiers from combat.^{[184][185]} Both militaries have made rapid progress in drone innovation whereby new developments happen in weeks instead of years due to the frequent interaction between military end users and defense contractors shortening feedback cycles. Reverse-engineering drone designs and strategies are also commonplace as shown by examples like Ukraine copying Russia's Molniya fixed-wing drones and Russia employing drone motherships equipped with Starlink.^{[186][187]}

The 2026 Iran war saw the crucial need for multiple countries to reevaluate their anti-drone capabilities in the wake of multiple missile and drone strikes that successfully hit military and civilian targets like airports, ports, and energy infrastructure throughout the Persian Gulf. During the war, Iran used thousands of mass-produced cheap drones like the Shahed-136 and missiles like the Fateh-110 which turned the Straits of Hormuz into a shooting gallery when Western air-defense missile stocks ran low. The costs of neutralizing a Shahed-136 was estimated at US\$20,000 to US\$50,000 compared to the expensive PATRIOT, THAAD, and Arrow-3 interceptor missiles which range from US\$3 to US\$12 million.^{[188][189]}



In the Russo-Ukrainian war, use of Russian UAVs increased about tenfold from early 2024 through summer 2025.^[167]



A Baykar Bayraktar TB2 of the Ukrainian Air Force armed with MAM-L; two ground control stations in the background

The war created a surge in demand for Ukrainian interceptor drones and anti-drone expertise in the Gulf countries, Israel, Europe, and the United States.^[190] FPV-based interceptors like the P1-Sun, STING, JEDI, Octopus, and Shvidun can reach speeds of up to 300 km/h and cost as low as around US\$1,000, thereby providing a cost-effective anti-drone solution as part of a multi-layered air defense network.

Civil applications

The civilian (commercial and general) drone market is dominated by Chinese companies. Chinese manufacturer DJI alone had 74% of the civil market share in 2018,^[191] with no other company accounting for more than 5%.^[192] The companies continue to hold over 70% of global market share by 2023, despite being under increasing scrutiny and sanctions from the United States.^[193] The US Interior Department grounded its fleet of DJI drones in 2020, while the Justice Department prohibited the use of federal funds for the purchase of DJI and other foreign-made UAVs.^{[194][195]} DJI is followed by American company 3D Robotics, Chinese company Yuneec, Autel Robotics, and French company Parrot.^{[192][196]} In 2025, Chinese drone companies held 90% of the global UAV market share, with DJI accounting 80 percent of the world market.^{[197][198]}

As of May 2021, 873,576 UAVs had been registered with the US FAA, of which 42% were categorized as commercial and 58% as recreational.^[199] 2018 NPD point to consumers increasingly purchasing drones with more advanced features with 33 percent growth in both the \$500+ and \$1000+ market segments.^[200]

The civil UAV market is relatively new compared to the military one. Companies are emerging in both developed and developing nations at the same time. Many early-stage startups have received support and funding from investors, as is the case in the United States, and from government agencies, as is the case in India.^[201] Some universities offer research and training programs or degrees.^[202] Private entities also provide online and in-person training programs for both recreational and commercial UAV use.^[203]

Consumer drones are widely used by police and military organizations worldwide because of the cost-effective nature of consumer products. Since 2018, the Israeli military have used DJI UAVs for light reconnaissance missions.^{[204][205][180]} DJI drones have been used by Chinese police in Xinjiang since 2017^{[206][207]} and American police departments nationwide since 2018.^{[208][209]} Both Ukraine and Russia used commercial DJI drones extensively during the Russian invasion of Ukraine.^[99] These civilian DJI drones were sourced by governments, hobbyists, and international donations to Ukraine and Russia to



3D design of a UAV high-altitude balloon launch^[168]



Zipline's aircraft being launched from a base in Rwanda to deliver blood products

support each side on the battlefield. They were often flown by drone hobbyists recruited by the armed forces. The prevalence of DJI drones was attributable to their market dominance, affordability, high performance, and reliability.^{[100][210]}

Aerial photography

Drones are ideally suited to capturing aerial shots in photography and cinematography, and are widely used for this purpose.^[166] Small drones avoid the need for precise coordination between pilot and cameraman, with the same person taking on both roles. Larger drones equipped with professional cinema cameras usually have a drone pilot and a camera operator who controls the camera angle and lens. For example, the AERIGON cinema drone, used in film production, is operated by two people.^[211] Drones provide access to dangerous, remote, or otherwise inaccessible sites.

Agriculture, forestry and environmental studies

As global demand for food production grows exponentially, resources are depleted, farmland is reduced, and agricultural labor is increasingly in short supply, there is an urgent need for more convenient and smarter agricultural solutions than traditional methods, and the agricultural drone and robotics industry is expected to make progress.^[212] Agricultural drones have been used to help build sustainable agriculture all over the world leading to a new generation of agriculture.^[213] In this context, there is a proliferation of innovations in both tools and methodologies which allow precise description of vegetation state and also may help to precisely distribute nutrients, pesticides or seeds over a field.^[7]

The use of UAVs is also being investigated to help detect and fight wildfires, whether through observation or launching pyrotechnic devices to start backfires.^[214]

UAVs are also now widely used to survey wildlife such as nesting seabirds, seals and even wombat burrows.^[215]



Agricultural drone on trailer setup



Entertainment

Drones are also used in nighttime displays for artistic and advertising purposes, with the main benefits being that they are safer, quieter, and better for the environment than fireworks. They can replace or be an adjunct to fireworks displays to reduce the financial burden of festivals. In addition they can complement fireworks due to the ability for drones to carry them, creating new forms of artwork in the process.^{[216][217][218]}

Drones can also be used for racing, either with or without VR functionality.

Environmental monitoring

UASs or UAVs offer a great advantage for environmental monitoring to generate a new generation of survey at very-high or ultra-high resolution both in space and time.^[219] This enables integration between satellites data and field-based monitoring. This has stimulated a huge number of activities in order to enhance the description of natural and agricultural ecosystems. Most common applications are:

- Topographic surveys^[220] for the production of orthomosaics, digital surface models and 3D models;
- Monitoring of natural ecosystems for biodiversity monitoring,^[221] habitat mapping,^[222] detection of invasive alien species^[223] and study of ecosystem degradation due to invasive species or disturbances;
- Precision agriculture^[224] which exploits all available technologies including UAV in order to produce more with less (e.g., optimisation of fertilizers, pesticides, irrigation);
- River monitoring several methods have been developed to perform flow monitoring using image velocimetry methods which allow to properly describe the 2D flow velocity fields.^[225]
- Structural integrity of any type of structure whether it be a dam, railway or other dangerous, inaccessible or massive locations for building monitoring.^[226]
- Mineral detection for acid mine drainage using UAVs and hyperspectral cameras can produce detailed maps of proxy minerals (e.g. goethite, jarosite) for certain pH-values in natural, mining and post-mining environments, such as remediated sites.^{[227][228]}
- In-situ gas measurements in potentially hazardous places like volcanic plumes.^[229]

These activities can be completed with different measurements, such as photogrammetry, thermography, multispectral images, 3D field scanning, gas detectors, and normalized difference vegetation index maps.

Geological hazards

UAVs have become a widely used tool for studying geohazards such as landslides.^[230] Various sensors, including radar, optical, and thermal, can be mounted on UAVs to monitor different properties. UAVs enable the capture of images of various landslide features, such as transverse, radial, and longitudinal cracks, ridges, scarps, and surfaces of rupture, even in inaccessible areas of the sliding mass.^{[231][232]} Moreover, processing the optical images captured by UAVs also allows for the creation of point clouds and 3D models, from which these properties can be derived.^[233] Comparing point clouds obtained at different times allows for the detection of changes caused by landslide deformation.^{[234][235]}

Humanitarian aid

Drones are increasingly finding their application in humanitarian aid and disaster relief, where they are used for a wide range of applications such as delivering food, medicine and essential items to remote areas or image mapping before and following disasters.^[236]

Law enforcement

Police can use drones for applications such as search and rescue and traffic monitoring.^[237]

Mineral exploration

UAVs may help in the discovery of new or reevaluation of known mineral deposits to meet the demand for raw materials such as critical raw metals (e.g. cobalt, nickel), rare earths and battery minerals. By employing a suite of sensors (e.g. spectral imaging, Lidar, magnetics, gamma-ray spectroscopy),^{[238][239]} and similar to those used in environmental monitoring, UAV-based data can produce maps of geological surface and subsurface features, contributing to more efficient and targeted mineral exploration.^{[240][241]}

Safety and security

Threats

Nuisance

UAVs can threaten airspace security in numerous ways, including unintentional collisions or other interference with other aircraft, deliberate attacks or by distracting pilots or flight controllers. The first incident of a drone-airplane collision occurred in mid-October 2017 in Quebec City, Canada.^[242] The first recorded instance of a drone collision with a hot air balloon occurred on 10 August 2018 in Driggs, Idaho, United States; although there was no significant damage to the balloon nor any injuries to its three occupants, the balloon pilot reported the incident to the National Transportation Safety Board, stating that "I hope this incident helps create a conversation of respect for nature, the airspace, and rules and regulations".^[243] Unauthorized UAV flights into or near major airports have prompted extended shutdowns of commercial flights.^[244]



US Department of Agriculture poster warning about the risks of flying UAVs near wildfires

Drones caused significant disruption at Gatwick Airport during December 2018, needing the deployment of the British Army.^{[245][246]}

In the United States, flying close to a wildfire is punishable by a maximum \$25,000 fine. Nonetheless, in 2014 and 2015, firefighting air support in California was hindered on several occasions, including at the Lake Fire^[247] and the North Fire.^{[248][249]} In response, California legislators introduced a bill that would allow firefighters to disable UAVs which invaded restricted airspace.^[250] The FAA later required registration of most UAVs.

Security vulnerabilities

By 2017, drones were being used to drop contraband into prisons.^[251]

The interest in UAVs cybersecurity has been raised greatly after the Predator UAV video stream hijacking incident in 2009,^[252] where Islamic militants used cheap, off-the-shelf equipment to stream video feeds from a UAV. Another risk is the possibility of hijacking or jamming a UAV in flight. Several security researchers have made public some vulnerabilities in commercial UAVs, in some cases even providing

full source code or tools to reproduce their attacks.^[253] At a workshop on UAVs and privacy in October 2016, researchers from the Federal Trade Commission showed they were able to hack into three different consumer quadcopters and noted that UAV manufacturers can make their UAVs more secure by the basic security measures of encrypting the Wi-Fi signal and adding password protection.^[254]

Privacy

A challenging issue is handling aerial drone technology within privacy laws.^[255]

Aggression

Many UAVs have been loaded with dangerous payloads, and/or crashed into targets. Payloads have included or could include explosives, chemical, radiological or biological hazards. UAVs with generally non-lethal payloads could possibly be hacked and put to malicious purposes. Counter-UAV systems (C-UAS), from detection to electronic warfare to UAVs designed to destroy other UAVs, are in development and being deployed by states to counter this threat.

Such developments have occurred despite the difficulties. As J. Rogers stated in a 2017 interview to A&T, "There is a big debate out there at the moment about what the best way is to counter these small UAVs, whether they are used by hobbyists causing a bit of a nuisance or in a more sinister manner by a terrorist actor".^[256]

Countermeasures

Counter unmanned air system

The malicious use of UAVs has led to the development of counter unmanned air system (C-UAS) technologies. Automatic tracking and detection of UAVs from commercial cameras have become accurate thanks to the development of deep learning based machine learning algorithms.^[257] It is also possible to automatically identify UAVs across different cameras with different viewpoints and hardware specification with re-identification methods.^[258] Commercial systems such as the Aaronia AARTOS have been installed on major international airports.^{[259][260]} Once a UAV is detected, it can be countered with kinetic force (missiles, projectiles or another UAV) or by non-kinetic force (laser, microwaves, communications jamming).^[261] Anti-aircraft missile systems such as the Iron Dome are also being enhanced with C-UAS technologies. Utilising a smart UAV swarm to counter one or more hostile UAVs is also proposed.^[262]



Italian Army soldiers of the 17th Anti-aircraft Artillery Regiment "Sforzesca" with a portable drone jammer in Rome

A variety of counter-UAS (C-UAS) systems have been developed globally to address the growing threat of small and tactical UAVs. These include multi-layered approaches combining radar, electro-optical sensors, radio frequency detection, and jamming technologies. Among them are systems developed by Elbit Systems, such as the ReDrone™ suite,^{[263][264]} which includes both fixed and portable configurations for detection and mitigation of drones in civilian and military environments. The Red Sky

2 system, also developed by Elbit, integrates multiple sensors and effectors and is designed to protect strategic sites from low-altitude aerial threats.^[265] Other developers, such as AirSight, specialize in the software and systems integration layer of C-UAS, utilizing AI and machine learning to analyze sensor data for drone detection and threat assessment in complex environments, as demonstrated by the launch of their AirGuard V3.18 system.^[129]



Cannon anti-drone system

Regulation

Regulatory bodies around the world are developing unmanned aircraft system traffic management solutions to better integrate UAVs into airspace.^[266]

The use of unmanned aerial vehicles is becoming increasingly regulated by the civil aviation authorities of individual countries. Regulatory regimes can differ significantly according to drone size and use. The International Civil Aviation Organization (ICAO) began exploring the use of drone technology as far back as 2005, which resulted in a 2011 report.^[267] France was among the first countries to set a national framework based on this report and larger aviation bodies such as the FAA and the EASA quickly followed suit.^[268] In 2021, the FAA published a rule requiring all commercially used UAVs and all UAVs regardless of intent weighing 250 g or more to participate in Remote ID, which makes drone locations, controller locations, and other information public from takeoff to shutdown; this rule has since been challenged in the pending federal lawsuit RaceDayQuads v. FAA.^{[269][270]}

In April 2026, China has introduced strict rules for recreational and civilian drones, require registration, IDs, real name, permit and real time monitoring. Beijing enacted near a total ban, limiting sales, transport and flights within the city. The regulation aim to ensure safety, manage low altitude airspace and address security.^[271]

EU Drone Certification – Class Identification Label

The implementation of the Class Identification Label (<https://eudroneport.com/blog/uas-class-label/>) serves a crucial purpose in the regulation and operation of drones.^[272] The label is a verification mechanism designed to confirm that drones within a specific class meet the rigorous standards set by administrations for design and manufacturing.^[273] These standards are necessary to ensure the safety and reliability of drones in various industries and applications.

By providing this assurance to customers, the Class Identification Label helps to increase confidence in drone technology and encourages wider adoption across industries. This, in turn, contributes to the growth and development of the drone industry and supports the integration of drones into society.

Export controls

The export of UAVs or technology capable of carrying a 500 kg payload at least 300 km is restricted in many countries by the Missile Technology Control Regime.

See also

- [Drone in a box](#)
- [Fiber optic drone](#)
- [Glide bomb](#)
- [International Aerial Robotics Competition](#)
- [List of films featuring drones](#)
- [List of military electronics of the United States](#)
- [List of unmanned aerial vehicles](#)
- [MARSS Interceptor](#)
- [Micromechanical Flying Insect](#)
- [Optionally piloted vehicle](#)
- [Sypaq Corvo Precision Payload Delivery System](#)
- [Satellite Sentinel Project](#)
- [Tactical Control System](#)
- [Unmanned underwater vehicle](#)

References

Notes

- a. Ukrainian: Сили безпілотних систем, СБС, *romanized: Syly bezpilotnykh system, SBS*, *lit. "Pilotless Systems Forces"*.
Russian: Войска беспилотных систем [ВБС], *romanized: Voyska bespilotnykh sistem [VBS]*, *lit. "Pilotless Systems Troops"*.

Citations

1. Gettinger, Daniel (2024). "3 Defining Drones". In Patton Rogers, James (ed.). *De Gruyter Handbook of Drone Warfare*. Berlin, Boston: De Gruyter. pp. 25–40. doi:10.1515/9783110742039-003 (<https://doi.org/10.1515%2F9783110742039-003>). ISBN 978-3-11-074203-9.
2. Pan, Honghao; Zahmatkesh, Mohsen; Rekabi-Bana, Fatemeh; Arvin, Farshad; Hu, Junyan (August 2025). "T-STAR: Time-Optimal Swarm Trajectory Planning for Quadrotor Unmanned Aerial Vehicles". *IEEE Transactions on Intelligent Transportation Systems*. **26** (8): 12532–12547. Bibcode:2025ITITr..2612532P (<https://ui.adsabs.harvard.edu/abs/2025ITITr..2612532P>). doi:10.1109/TITS.2025.3557783 (<https://doi.org/10.1109%2FTITS.2025.3557783>).
3. Tice, Brian P. (Spring 1991). "Unmanned Aerial Vehicles – The Force Multiplier of the 1990s" (<https://web.archive.org/web/20090724015052/http://www.airpower.maxwell.af.mil/airchronicles/apj/apj91/spr91/4spr91.htm>). *Airpower Journal*. Archived from the original (<http://www.airpower.maxwell.af.mil/airchronicles/apj/apj91/spr91/4spr91.htm>) on 24 July 2009. Retrieved 6 June 2013. "When used, UAVs should generally perform missions characterized by the three Ds: dull, dirty, and dangerous."
4. Alvarado, Ed (3 May 2021). "237 Ways Drone Applications Revolutionize Business" (<https://droneii.com/237-ways-drone-applications-revolutionize-business>). *Drone Industry Insights*. Archived (<https://web.archive.org/web/20210511113408/https://droneii.com/237-ways-drone-applications-revolutionize-business>) from the original on 11 May 2021. Retrieved 11 May 2021.
5. Rekabi-Bana, Fatemeh; Hu, Junyan; Krajník, Tomáš; Arvin, Farshad (March 2024). "Unified Robust Path Planning and Optimal Trajectory Generation for Efficient 3D Area Coverage of Quadrotor UAVs" (<https://durham-repository.worktribe.com/output/1899538>). *IEEE Transactions on Intelligent Transportation Systems*. **25** (3): 2492–2507. Bibcode:2024ITITr..25.2492R (<https://ui.adsabs.harvard.edu/abs/2024ITITr..25.2492R>). doi:10.1109/TITS.2023.3320049 (<https://doi.org/10.1109%2FTITS.2023.3320049>).

6. Hu, Junyan; Niu, Hanlin; Carrasco, Joaquin; Lennox, Barry; Arvin, Farshad (April 2022). "Fault-tolerant cooperative navigation of networked UAV swarms for forest fire monitoring". *Aerospace Science and Technology*. **123** 107494. Bibcode:2022AeST..12307494H (<https://ui.adsabs.harvard.edu/abs/2022AeST..12307494H>). doi:10.1016/j.ast.2022.107494 (<https://doi.org/10.1016%2Fj.ast.2022.107494>).
7. Manfreda, Salvatore; Ben Dor, Eyal (2023). "Remote sensing of the environment using unmanned aerial systems". *Unmanned Aerial Systems for Monitoring Soil, Vegetation, and Riverine Environments*. pp. 3–36. doi:10.1016/B978-0-323-85283-8.00009-6 (<https://doi.org/10.1016%2FB978-0-323-85283-8.00009-6>). ISBN 978-0-323-85283-8.
8. Perks, Matthew T.; Dal Sasso, Silvano Fortunato; Hauet, Alexandre; Jamieson, Elizabeth; Le Coz, Jérôme; Pearce, Sophie; Peña-Haro, Salvador; Pizarro, Alonso; Strelnikova, Dariia; Tauro, Flavia; Bomhof, James; Grimaldi, Salvatore; Goulet, Alain; Hortobágyi, Borbála; Jodeau, Magali; Käfer, Sabine; Ljubičić, Robert; Maddock, Ian; Mayr, Peter; Paulus, Gernot; Pénard, Lionel; Sinclair, Leigh; Manfreda, Salvatore (8 July 2020). "Towards harmonisation of image velocimetry techniques for river surface velocity observations" (<https://doi.org/10.5194%2Fessd-12-1545-2020>). *Earth System Science Data*. **12** (3): 1545–1559. Bibcode:2020ESSD...12.1545P (<https://ui.adsabs.harvard.edu/abs/2020ESSD...12.1545P>). doi:10.5194/essd-12-1545-2020 (<https://doi.org/10.5194%2Fessd-12-1545-2020>). hdl:11563/161555 (<https://hdl.handle.net/11563%2F161555>).
9. Koparan, Cengiz; Koc, A. Bulent; Privette, Charles V.; Sawyer, Calvin B. (March 2020). "Adaptive Water Sampling Device for Aerial Robots" (<https://doi.org/10.3390%2Fdrones4010005>). *Drones*. **4** (1): 5. Bibcode:2020Drone...4....5K (<https://ui.adsabs.harvard.edu/abs/2020Drone...4....5K>). doi:10.3390/drones4010005 (<https://doi.org/10.3390%2Fdrones4010005>).
10. Koparan, Cengiz; Koc, Ali Bulent; Privette, Charles V.; Sawyer, Calvin B.; Sharp, Julia L. (May 2018). "Evaluation of a UAV-Assisted Autonomous Water Sampling" (<https://doi.org/10.3390%2Fw10050655>). *Water*. **10** (5): 655. Bibcode:2018Water..10..655K (<https://ui.adsabs.harvard.edu/abs/2018Water..10..655K>). doi:10.3390/w10050655 (<https://doi.org/10.3390%2Fw10050655>).
11. Koparan, Cengiz; Koc, Ali Bulent; Privette, Charles V.; Sawyer, Calvin B. (March 2018). "In Situ Water Quality Measurements Using an Unmanned Aerial Vehicle (UAV) System" (<https://doi.org/10.3390%2Fw10030264>). *Water*. **10** (3): 264. Bibcode:2018Water..10..264K (<https://ui.adsabs.harvard.edu/abs/2018Water..10..264K>). doi:10.3390/w10030264 (<https://doi.org/10.3390%2Fw10030264>).
12. Koparan, Cengiz; Koc, Ali Bulent; Privette, Charles V.; Sawyer, Calvin B. (March 2019). "Autonomous In Situ Measurements of Noncontaminant Water Quality Indicators and Sample Collection with a UAV" (<https://doi.org/10.3390%2Fw11030604>). *Water*. **11** (3): 604. Bibcode:2019Water..11..604K (<https://ui.adsabs.harvard.edu/abs/2019Water..11..604K>). doi:10.3390/w11030604 (<https://doi.org/10.3390%2Fw11030604>).
13. "Drones smuggling porn, drugs to inmates around the world" (<https://www.foxnews.com/us/drones-smuggling-porn-drugs-to-inmates-around-the-world/>). *Fox News*. 17 April 2017. Archived (<https://web.archive.org/web/20180831035314/http://www.foxnews.com/us/2017/04/17/drones-smuggling-porn-drugs-to-inmates-around-world.html>) from the original on 31 August 2018. Retrieved 17 April 2017.
14. "unmanned aerial vehicle" (<http://www.thefreedictionary.com/Unmanned+Aerial+Vehicle>). *TheFreeDictionary.com*. Archived (<https://web.archive.org/web/20150108233752/http://www.thefreedictionary.com/Unmanned+Aerial+Vehicle>) from the original on 8 January 2015. Retrieved 8 January 2015.
15. Guilmartin, John F. "unmanned aerial vehicle" (<https://www.britannica.com/technology/unmanned-aerial-vehicle>). *Encyclopedia Britannica*. Archived (<https://web.archive.org/web/20200329092652/https://www.britannica.com/technology/unmanned-aerial-vehicle>) from the original on 29 March 2020. Retrieved 24 March 2020.

16. "14 CFR Part 107 (Docket FAA-2015-0150, Amdt. 107-1, 81 FR 42209, June 28, 2016, as amended by Amdt. No. 107-8, 86 FR 4381, Jan. 15, 2021)" (<https://www.ecfr.gov/current/title-14/part-107>). *eCFR :: 14 CFR Part 107 -- Small Unmanned Aircraft Systems (FAR Part 107)*. Federal Aviation Administration, Department of Transportation. 15 January 2021. Retrieved 16 February 2025.
17. "Unmanned Aircraft Systems Roadmap" (<https://web.archive.org/web/20081002220516/http://www.acq.osd.mil/usd/Roadmap%20Final2.pdf#search=%22Dod%20UAS%20Roadmap%202005%22>) (PDF). Archived from the original (<http://www.acq.osd.mil/usd/Roadmap%20Final2.pdf#search=%22Dod%20UAS%20Roadmap%202005%22>) (PDF) on 2 October 2008.
18. "European ATM Master Plan 2015 | SESAR" (<https://web.archive.org/web/20160206104147/http://www.sesarju.eu/newsroom/brochures-publications/european-atm-master-plan-2015>). *www.sesarju.eu*. Archived from the original (<https://www.sesarju.eu/newsroom/brochures-publications/european-atm-master-plan-2015>) on 6 February 2016. Retrieved 3 February 2016.
19. "State government gears up for autonomous RPAS mapping" (<http://www.spatialsource.com.au/unmanned/state-government-gears-autonomous-rpas-mapping>). 23 January 2017. Archived (<https://web.archive.org/web/20170225064802/http://www.spatialsource.com.au/unmanned/state-government-gears-autonomous-rpas-mapping>) from the original on 25 February 2017. Retrieved 1 February 2017.
20. "Canadian Aviation Regulations" (<https://lois-laws.justice.gc.ca/eng/regulations/SOR-96-433/FullText.html#s-900.01>). *Government of Canada – Justice Laws Website*. 1 June 2019. Archived (<https://web.archive.org/web/20220106213439/https://lois-laws.justice.gc.ca/eng/regulations/SOR-96-433/FullText.html#s-900.01>) from the original on 6 January 2022. Retrieved 16 January 2019.
21. Note; the term "drone" refers to the male bee that serves only to fertilize the queen bee, hence the use of the name in reference to the DH Queen Bee aerial target.
22. "The FAA is shifting from unmanned to uncrewed" (<https://skymantics.com/the-faa-is-shifting-from-unmanned-to-uncrewed-why-is-it-important/>). 7 June 2023.
23. "Defence Drone Strategy - the UK's approach to Defence Uncrewed Systems" (<https://www.gov.uk/government/publications/defence-drone-strategy-the-uks-approach-to-defence-uncrewed-systems/defence-drone-strategy-the-uks-approach-to-defence-uncrewed-systems>). GOV.UK.
24. "Aviation Management: Chapter 12 - Uncrewed Aircraft Systems" (https://www.nps.gov/subjects/aviation/upload/RM60_Chapter-12_Uncrewed-Aircraft-Systems-508-signed.pdf) (PDF).
25. "Drones and Artificial Intelligence" (<https://www.droneii.com/drones-and-artificial-intelligence>). *Drone Industry Insights*. 28 August 2018. Archived (<https://web.archive.org/web/20200917052156/https://www.droneii.com/drones-and-artificial-intelligence>) from the original on 17 September 2020. Retrieved 11 April 2020.
26. "What is the difference between a drone and an RC plane or helicopter?" (<https://web.archive.org/web/20151117020459/https://www.dronesetc.com/blogs/news/21481409-what-is-the-difference-between-a-drone-and-an-rc-plane-or-helicopter>). *Drones Etc.* Drones Etc. Archived from the original (<https://www.dronesetc.com/blogs/news/21481409-what-is-the-difference-between-a-drone-and-an-rc-plane-or-helicopter>) on 17 November 2015. Retrieved 12 October 2015.
27. "UAV classification" (<https://www.e-education.psu.edu/geog892/node/5>). Archived (<https://web.archive.org/web/20220523235132/https://www.e-education.psu.edu/geog892/node/5>) from the original on 23 May 2022. Retrieved 10 June 2022.
28. "Eyes of the Army: U.S. Army Roadmap for UAS 2010–2035" (<https://irp.fas.org/program/collect/uas-army.pdf>) (PDF). Archived (<https://web.archive.org/web/20220218054255/https://irp.fas.org/program/collect/uas-army.pdf>) (PDF) from the original on 18 February 2022. Retrieved 10 June 2022.

29. "Nano, micro, small: The different drone types in India & if Jammu-like strike can be averted" (<https://theprint.in/defence/nano-micro-small-the-different-drone-types-in-india-if-jammu-like-strike-can-be-averted/686158/>). Archived (<https://web.archive.org/web/20210629111522/http://theprint.in/defence/nano-micro-small-the-different-drone-types-in-india-if-jammu-like-strike-can-be-averted/686158/>) 29 June 2021 at the Wayback Machine. *ThePrint*, 29 June 2021.
30. Drones, Percepto (3 January 2019). "The Differences Between UAV, UAS, and Autonomous Drones" (<https://percepto.co/what-are-the-differences-between-uav-uas-and-autonomous-drones/>). *Percepto*. Archived (<https://web.archive.org/web/20200218072255/https://percepto.co/what-are-the-differences-between-uav-uas-and-autonomous-drones/>) from the original on 18 February 2020. Retrieved 18 February 2020.
31. Cary, Leslie; Coyne, James. "ICAO Unmanned Aircraft Systems (UAS), Circular 328". *2011–2012 UAS Yearbook – UAS: The Global Perspective* (https://web.archive.org/web/20160304025432/http://uvs-info.com/phocadownload/05_3a_2011/P112-P115_C%26AI_ICAO-Advisory-Circular.pdf) (PDF). Blyenburgh & Co. pp. 112–115. Archived from the original (http://uvs-info.com/phocadownload/05_3a_2011/P112-P115_C&AI_ICAO-Advisory-Circular.pdf) (PDF) on 4 March 2016. Retrieved 26 February 2022.
32. Hu, Junyan; Lanzon, Alexander (May 2018). "An innovative tri-rotor drone and associated distributed aerial drone swarm control" ([https://www.research.manchester.ac.uk/portal/en/publications/an-innovative-trirotor-drone-and-associated-distributed-aerial-drone-swarm-control\(4b61945e-f961-44c9-8b22-ad644b49f726\).html](https://www.research.manchester.ac.uk/portal/en/publications/an-innovative-trirotor-drone-and-associated-distributed-aerial-drone-swarm-control(4b61945e-f961-44c9-8b22-ad644b49f726).html)). *Robotics and Autonomous Systems*. **103**: 162–174. doi:10.1016/j.robot.2018.02.019 (<https://doi.org/10.1016%2Fj.robot.2018.02.019>).
33. Garrow, Laurie A.; German, Brian J.; Leonard, Caroline E. (1 November 2021). "Urban air mobility: A comprehensive review and comparative analysis with autonomous and electric ground transportation for informing future research" (<https://doi.org/10.1016%2Fj.trc.2021.103377>). *Transportation Research Part C: Emerging Technologies*. **132** 103377. Bibcode:2021TRPC..13203377G (<https://ui.adsabs.harvard.edu/abs/2021TRPC..13203377G>). doi:10.1016/j.trc.2021.103377 (<https://doi.org/10.1016%2Fj.trc.2021.103377>).
34. "Exploring Gas Powered Drones: Uses and Benefits" (<https://www.flyability.com/blog/gas-powered-drone>). *www.flyability.com*. Retrieved 8 August 2024.
35. Zhang, Caizhi; Qiu, Yuqi; Chen, Jiawei; Li, Yuehua; Liu, Zhitao; Liu, Yang; Zhang, JiuJun; Hwa, Chan Siew (August 2022). "A comprehensive review of electrochemical hybrid power supply systems and intelligent energy managements for unmanned aerial vehicles in public services". *Energy and AI*. **9** 100175. Bibcode:2022EneAI...900175Z (<https://ui.adsabs.harvard.edu/abs/2022EneAI...900175Z>). doi:10.1016/j.egyai.2022.100175 (<https://doi.org/10.1016%2Fj.egyai.2022.100175>). hdl:10356/164036 (<https://hdl.handle.net/10356%2F164036>).
36. Nefedkin, Sergey I.; Klimova, Mariya A.; Glasov, Vasily S.; Pavlov, Vladislav I.; Tolmachev, Yuriy V. (June 2021). "Effect of the corrugated bipolar plate design on the self-humidification of a high power density PEMFC stack for UAVs". *Fuel Cells*. **21** (3): 234–253. doi:10.1002/fuce.202000163 (<https://doi.org/10.1002%2Fuce.202000163>).
37. "Powering Solutions for Your Drone in 2024: New Fuels" (<https://www.commercialuavnews.com/surveying/powering-solutions-for-your-drone-in-2024-new-fuels>). *Commercial Uav News*. Retrieved 8 August 2024.
38. Tabibi, Alex (30 January 2024). "Solar-Powered Drones and UAVs" (<https://green.org/2024/01/30/solar-powered-drones-and-uavs/>). *Green.org*. Retrieved 28 October 2025.
39. Fabled Sky Research (2024). Revolutionizing UAV Capabilities: Exploring the Potential of Nuclear Propulsion Systems (Preprint). doi:10.6084/m9.figshare.26198462 (<https://doi.org/10.6084%2Fm9.figshare.26198462>).
40. The *Encyclopedia of the Arab-Israeli Conflict: A Political, Social, and Military History: A Political, Social, and Military History*, ABC-CLIO, 12 May 2008, by Spencer C. Tucker, Priscilla Mary Roberts, pages 1054–55 ISBN

41. The Future of Drone Use: Opportunities and Threats from Ethical and Legal Perspectives (<https://books.google.com/books?id=YSSPAgAAQBAJ&pg=PT43>) Archived (<https://web.archive.org/web/20230227213802/https://books.google.com/books?id=YSSPAgAAQBAJ&pg=PT43>) 27 February 2023 at the [Wayback Machine](#), Asser Press – Springer, chapter by Alan McKenna, page 355
42. Kaplan, Philip (2013). *Naval Aviation in the Second World War* (<https://books.google.com/books?id=pDARBQAAQBAJ&pg=PT19>). Pen and Sword. p. 19. ISBN 978-1-4738-2997-8. Archived (<https://web.archive.org/web/20230227213802/https://books.google.com/books?id=pDARBQAAQBAJ&pg=PT19>) from the original on 27 February 2023. Retrieved 19 August 2019.
43. Hallion, Richard P. (2003). *Taking Flight: Inventing the Aerial Age, from Antiquity through the First World War* (<https://archive.org/details/takingflightinve0000hall>). Oxford University Press. p. 66 (<https://archive.org/details/takingflightinve0000hall/page/66>). ISBN 978-0-19-028959-1.
44. Naval Aviation in the First World War: Its Impact and Influence (<https://archive.org/details/navalaviationinf00laym>), R. D. Layman, page 56
45. Renner, Stephen L. (2016). *Broken Wings: The Hungarian Air Force, 1918–45* (<https://books.google.com/books?id=HDoJDgAAQBAJ&pg=PP1>). Indiana University Press. p. 2. ISBN 978-0-253-02339-1. Archived (<https://web.archive.org/web/20230227213803/https://books.google.com/books?id=HDoJDgAAQBAJ&pg=PP1>) from the original on 27 February 2023. Retrieved 26 October 2019.
46. Murphy, Justin D. (2005). *Military Aircraft, Origins to 1918: An Illustrated History of Their Impact* (<https://books.google.com/books?id=7pS1QpH8FRgC&pg=PA9>). ABC-CLIO. pp. 9–10. ISBN 978-1-85109-488-2. Archived (<https://web.archive.org/web/20230227213804/https://books.google.com/books?id=7pS1QpH8FRgC&pg=PA9>) from the original on 27 February 2023. Retrieved 19 August 2019.
47. Haydon, F. Stansbury (2000). *Military Ballooning During the Early Civil War* (<https://archive.org/details/militaryballooni00hayd>). JHU Press. pp. 18 (<https://archive.org/details/militaryballooni00hayd/page/18>)–20. ISBN 978-0-8018-6442-1.
48. Mikesch, Robert C. (1973). "Japan's World War II balloon bomb attacks on North America". *Smithsonian Annals of Flight* (9): 1–85. doi:10.5479/si.AnnalsFlight.9 (<https://doi.org/10.5479/si.AnnalsFlight.9>). hdl:10088/18679 (<https://hdl.handle.net/10088/18679>).
49. Tapan K. Sarkar, *History of wireless*, John Wiley and Sons, 2006, ISBN 0-471-71814-9, p. 97.
50. Biodiversity Heritage Library. *Mécanique Appliquée. - Sur le télékine. Note de M. L. Torres, présentée par M. Appell* (<https://www.biodiversitylibrary.org/item/31206#page/323/mode/1up>) 3 August 1903, pp. 317-319, Comptes rendus de l'Académie des Sciences.
51. Randy Alfred, "Nov. 7, 1905: Remote Control Wows Public (<https://www.wired.com/2011/11/1107wireless-remote-control/>)", *Wired*, 7 November 2011.
52. H. R. Everett (2015). *Unmanned Systems of World Wars I and II*. MIT Press. pp. 91–95. ISBN 978-0-262-02922-3.
53. Taylor, John W. R.. *Jane's Pocket Book of Remotely Piloted Vehicles*.
54. Professor A. M. Low FLIGHT, 3 October 1952 page 436 "The First Guided Missile"
55. Dempsey, Martin E. (9 April 2010). "Eyes of the Army—U.S. Army Roadmap for Unmanned Aircraft Systems 2010–2035" (<https://web.archive.org/web/20180922052536/http://www.rucker.army.mil/usaace/uas/US%20Army%20UAS%20RoadMap%202010%202035.pdf>) (PDF). U.S. Army. Archived from the original (<http://www.rucker.army.mil/usaace/uas/US%20Army%20UAS%20RoadMap%202010%202035.pdf>) (PDF) on 22 September 2018. Retrieved 6 March 2011.

56. Says, Robert Kanyike (21 May 2012). "History of U.S. Drones" (<http://understandingempire.wordpress.com/2-0-a-brief-history-of-u-s-drones/>). Archived (<https://web.archive.org/web/20190926020824/https://understandingempire.wordpress.com/2-0-a-brief-history-of-u-s-drones/>) from the original on 26 September 2019. Retrieved 17 February 2014.
57. Andersson, Lennart (1994). *Soviet Aircraft and Aviation, 1917–1941* (<https://books.google.com/books?id=1lzbAAAAMAAJ>). The Putnam Aviation Series. Annapolis, Maryland: Naval Institute Press. p. 249. ISBN 978-1-55750-770-9. Archived (<https://web.archive.org/web/20230227213805/https://books.google.com/books?id=1lzbAAAAMAAJ>) from the original on 27 February 2023. Retrieved 18 December 2021. "Experiments with a pilotless drone version of the TB-1 controlled by radio from other aircraft started in 1935 and continued until 1939."
58. H. R. Everett (2015). *Unmanned Systems of World Wars I and II*. MIT Press. p. 318 (<https://books.google.com/books?id=fNjgCgAAQBAJ&pg=PA318>). ISBN 978-0-262-02922-3.
59. Wagner 1982, p. xi.
60. Wagner 1982, p. xi, xii.
61. Wagner 1982, p. xii.
62. Wagner 1982, p. 79.
63. Wagner 1982, p. 78, 79.
64. Dunstan, Simon (2013). *Israeli Fortifications of the October War 1973* (<https://books.google.com/books?id=CPm8EoclfoQC&pg=PA16>). Osprey Publishing. p. 16. ISBN 978-1-78200-431-8. Retrieved 25 October 2015. "The War of Attrition was also notable for the first use of UAVs, or unmanned aerial vehicles, carrying reconnaissance cameras in combat."
65. Saxena, V. K. (2013). *The Amazing Growth and Journey of UAV's and Ballistic Missile Defence Capabilities: Where the Technology is Leading to?* (<https://books.google.com/books?id=hwWqCQAAQBAJ&pg=PA6>). Vij Books India Pvt Ltd. p. 6. ISBN 978-93-82573-80-7. Archived (<https://web.archive.org/web/20230227213814/https://books.google.com/books?id=hwWqCQAAQBAJ&pg=PA6>) from the original on 27 February 2023. Retrieved 25 October 2015. "During the Yom Kippur War the Israelis used Teledyne Ryan 124 R RPVs along with the home-grown Scout and Mastiff UAVs for reconnaissance, surveillance, and as decoys to draw fire from Arab SAMs. This resulted in Arab forces expending costly and scarce missiles on inappropriate targets [...]."
66. Blum, Howard (2003). *The eve of destruction: the untold story of the Yom Kippur War*. HarperCollins. ISBN 978-0-06-001399-8.
67. Wagner 1982, p. 202.
68. Wagner 1982, p. 200, 212.
69. Wagner 1982, p. 208.
70. "A Brief History of UAVs" (<http://www.howstuffworks.com/reaper1.htm>). Howstuffworks.com. 22 July 2008. Archived (<https://web.archive.org/web/20130522111255/http://www.howstuffworks.com/reaper1.htm>) from the original on 22 May 2013. Retrieved 8 January 2015.
71. "Russia Buys A Bunch of Israeli UAVs" (<http://www.strategypage.com/htmww/htairfo/articles/20090409.aspx>). Strategypage.com. Archived (<https://web.archive.org/web/20131026110814/http://strategypage.com/htmww/htairfo/articles/20090409.aspx>) from the original on 26 October 2013. Retrieved 8 January 2015.
72. Azoulai, Yuval (24 October 2011). "Unmanned combat vehicles shaping future warfare" (<http://www.globes.co.il/serveen/globes/docview.asp?did=1000691790>). Globes. Archived (<https://web.archive.org/web/20131203060756/http://www.globes.co.il/serveen/globes/docview.asp?did=1000691790>) from the original on 3 December 2013. Retrieved 8 January 2015.
73. Levinson, Charles (13 January 2010). "Israeli Robots Remake Battlefield" (<https://www.wsj.com/articles/SB126325146524725387>). *The Wall Street Journal*. p. A10. Archived (<https://web.archive.org/web/20200313163718/https://www.wsj.com/articles/SB126325146524725387>) from the original on 13 March 2020. Retrieved 13 January 2010.

74. Gal-Or, Benjamin (1990). *Vectored Propulsion, Supermaneuverability & Robot Aircraft*. Springer Verlag. ISBN 978-3-540-97161-0.
75. Fuller, Christopher J. (2015). "The Eagle Comes Home to Roost: The Historical Origins of the CIA's Lethal Drone Program". *Intelligence and National Security*. **30** (6): 769–792. doi:10.1080/02684527.2014.895569 (<https://doi.org/10.1080%2F02684527.2014.895569>).
76. Goraj, Z.; Frydrychewicz, A.; Świtkiewicz, R.; Hernik, B.; Gadomski, J.; Goetzendorf-Grabowski, T.; Figat, M.; Suchodolski, S. (2004). "High altitude long endurance unmanned aerial vehicle of a new generation - a design challenge for a low cost, reliable and high performance aircraft". *Bulletin of the Polish Academy of Sciences. Technical Sciences*. **52** (3): 173–194.
77. Community Research and Development Information Service. *Civil UAV application and economic effectiveness of potential configuration solutions* (http://cordis.europa.eu/project/rcn/63495_en.html). Publications Office of the European Union. Archived (https://web.archive.org/web/20160129123521/http://cordis.europa.eu/project/rcn/63495_en.html) from the original on 29 January 2016. Retrieved 9 December 2015.
78. Ackerman, Spencer; Shachtman, Noah (9 January 2012). "Almost 1 in 3 U.S. Warplanes Is a Robot" (<https://www.wired.com/2012/01/drone-report/>). *Wired*. Archived (<https://web.archive.org/web/20200323171457/https://www.wired.com/2012/01/drone-report/>) from the original on 23 March 2020. Retrieved 8 January 2015.
79. Singer, Peter W. "A Revolution Once More: Unmanned Systems and the Middle East" (http://www.brookings.edu/articles/2009/11_robotic_revolution_singer.aspx). Archived (https://web.archive.org/web/20110806065520/http://www.brookings.edu/articles/2009/11_robotic_revolution_singer.aspx) 6 August 2011 at the *Wayback Machine*. The Brookings Institution. November 2009.
80. Radsan, A. J.; Murphy, Richard (2011). "Measure Twice, Shoot Once: Higher Care for CIA Targeted Killing". *University of Illinois Law Review*. **2011** (4): 1201–1241. SSRN 1625829 (https://papers.ssrn.com/sol3/papers.cfm?abstract_id=1625829).
81. Sayler (2015)
82. Franke, Ulrike Esther (2015). "The global diffusion of unmanned aerial vehicles (UAVs) or 'drones'". In Mike Aaronson, ed. *Precision Strike Warfare and International Intervention*. Routledge.
83. "Who Invented the Drone: UAV History Lesson" (<https://dronelaunchacademy.com/resources/who-invented-the-drone-uav-history-lesson/>). *Drone Launch Academy*. 4 September 2024.
84. Santo, Brian (27 December 2018). "The Consumer Electronics Hall of Fame: DJI Phantom Drone" (<https://spectrum.ieee.org/the-consumer-electronics-hall-of-fame-dji-phantom-drone>). *IEEE Spectrum*.
85. Chiu, Karen (2 October 2018). "The story of drone pioneer DJI" (<https://www.scmp.com/abcus/who-what/what/article/3028241/story-drone-pioneer-dji>). *South China Morning Post*.
86. Bateman, Joshua (1 September 2017). "China drone maker DJI: Alone atop the unmanned skies" (<https://www.cnbc.com/2017/09/01/in-race-to-dominate-drone-space-west-is-no-match-for-chinas-dji.html>). *News Ledge*. Archived (<https://web.archive.org/web/20180219151200/https://www.cnbc.com/2017/09/01/in-race-to-dominate-drone-space-west-is-no-match-for-chinas-dji.html>) from the original on 19 February 2018. Retrieved 19 February 2018.
87. Hambling, David. "Drones may have attacked humans fully autonomously for the first time" (<https://www.newscientist.com/article/2278852-drones-may-have-attacked-humans-fully-autonomously-for-the-first-time/>). *New Scientist*. Archived (<https://web.archive.org/web/20210730173534/https://www.newscientist.com/article/2278852-drones-may-have-attacked-humans-fully-autonomously-for-the-first-time/>) from the original on 30 July 2021. Retrieved 30 May 2021.

88. Forestier-Walker, Robin (13 October 2020). "Nagorno-Karabakh: New weapons for an old conflict spell danger" (<https://www.aljazeera.com/features/2020/10/13/nagorno-karabakh-new-weapons-for-an-old-conflict-spell-danger>). *Al Jazeera*. Archived (<https://web.archive.org/web/20201013185319/https://www.aljazeera.com/features/2020/10/13/nagorno-karabakh-new-weapons-for-an-old-conflict-spell-danger>) from the original on 13 October 2020. Retrieved 18 December 2021. "... battlefield videos and the known military capabilities of the two warring sides suggest Azerbaijan has the technological advantage, especially with its combat drones purchased from Israel and Turkey."
89. name="IngenuityJPL">"Ingenuity Mars Helicopter" (<https://www.jpl.nasa.gov/missions/ingenuity/>). *NASA Jet Propulsion Laboratory*. California Institute of Technology. Retrieved 8 April 2026. "Ingenuity arrived at the Red Planet on February 18, 2021... On Jan. 18, 2024, the helicopter made its 72nd and final flight after damaging a rotor blade during landing."
90. Bailon-Ruiz, Rafael; Lacroix, Simon; Bit-Monnot, Arthur (2018). "Planning to Monitor Wildfires with a Fleet of UAVs". *2018 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)* (<https://web.archive.org/web/20221229172238/https://ieeexplore.ieee.org/document/8593859/>). IEEE. pp. 4729–4734. doi:10.1109/IROS.2018.8593859 (<https://doi.org/10.1109%2FIROS.2018.8593859>). ISBN 978-1-5386-8094-0. Archived from the original (<https://hal.laas.fr/hal-01852176>) on 29 December 2022. Retrieved 22 July 2023.
91. Williams, Dan; Hafezi, Parisa (14 April 2024). "Iran launches retaliatory attack on Israel with hundreds of drones, missiles" (<https://www.reuters.com/world/middle-east/iran-launches-drone-attack-israel-expected-unfold-over-hours-2024-04-13/>). Reuters. Retrieved 10 June 2025.
92. "Iran launches more than 300 drones and missiles at Israel; Biden condemns attack" (<https://www.washingtonpost.com/world/2024/04/13/iran-israel-hamas-war-news-gaza-palestine/>). *The Washington Post*.
93. "Iran launches hundreds of missiles, drones in first direct attack on Israel" (<https://www.aljazeera.com/news/2024/4/13/israeli-army-says-iran-has-launched-drones-at-israel/>). *Al Jazeera*. Retrieved 10 June 2025.
94. Williams, Dan (13 April 2024). "Iran launches drone attack at Israel, expected to unfold 'over hours'" (<https://www.reuters.com/world/middle-east/ukmto-receives-report-incident-50nm-northeast-uaes-fujairah-2024-04-13/>). Reuters. Retrieved 10 June 2025.
95. "The largest drone attack in history" (<http://iranpress.com/content/277652/the-largest-drone-attack-history>). *iranpress.com*. Retrieved 10 June 2025.
96. Hill, John (7 May 2024). "In data: UAS market projected to nearly double in ten years" (<https://www.army-technology.com/news/in-data-uas-market-projected-to-nearly-double-in-ten-years/>). *Army Technology*. Retrieved 8 May 2024.
97. Milasauskas, Tomas; Jaškūnas, Liudvikas (20 June 2024). "FPV drones in Ukraine are changing modern warfare" (<https://www.atlanticcouncil.org/blogs/ukrainealert/fpv-drones-in-ukraine-are-changing-modern-warfare/>). *Atlantic Council*.
98. Kunertova, Dominika (2023). "The war in Ukraine shows the game-changing effect of drones depends on the game". *Bulletin of the Atomic Scientists*. **79** (2): 95–102. Bibcode:2023BuAtS..79b..95K (<https://ui.adsabs.harvard.edu/abs/2023BuAtS..79b..95K>). doi:10.1080/00963402.2023.2178180 (<https://doi.org/10.1080%2F00963402.2023.2178180>).
99. Sangma, Mike (25 December 2022). "Ukraine has an unlikely ally in fight against Russia: DJI drones" (<https://www.eastmojo.com/recent-top-news/2022/12/25/ukraine-has-an-unlikely-ally-in-fight-against-russia-dji-drones/>). *East Mojo*. Archived (<https://web.archive.org/web/20230220104418/https://www.eastmojo.com/recent-top-news/2022/12/25/ukraine-has-an-unlikely-ally-in-fight-against-russia-dji-drones/>) from the original on 20 February 2023. Retrieved 26 December 2022.

100. Greenwood, Faine (16 February 2023). "The Drone War in Ukraine Is Cheap, Deadly, and Made in China" (<https://foreignpolicy.com/2023/02/16/ukraine-russia-war-drone-warfare-china/>). *Foreign Policy*. Retrieved 6 March 2023.
101. Mozur, Paul; Krolik, Aaron; Bradsher, Keith (21 March 2023). "As War in Ukraine Grinds On, China Helps Refill Russian Drone Supplies" (<https://www.nytimes.com/2023/03/21/business/russia-china-drones-ukraine-war.html>). *The New York Times*. Archived (<https://web.archive.org/web/20230602144130/https://www.nytimes.com/2023/03/21/business/russia-china-drones-ukraine-war.html>) from the original on 2 June 2023. Retrieved 23 March 2023.
102. Somerville, Heather; Forrest, Brett (10 April 2024). "How American Drones Failed to Turn the Tide in Ukraine" (<https://www.wsj.com/world/how-american-drones-failed-to-turn-the-tide-in-ukraine-b0ebbac3>). *The Wall Street Journal*. Archived (<https://web.archive.org/web/20240724030203/https://www.wsj.com/world/how-american-drones-failed-to-turn-the-tide-in-ukraine-b0ebbac3>) from the original on 24 July 2024. Retrieved 24 July 2024.
103. Baptista, Eduardo (28 March 2022). "China's DJI rejects claim that Russian military uses its drones in Ukraine" (<https://www.reuters.com/world/china/chinas-dji-rejects-claim-that-russian-military-uses-its-drones-ukraine-2022-03-28/>). *Reuters*. Archived (<https://web.archive.org/web/20220328082714/https://www.reuters.com/world/china/chinas-dji-rejects-claim-that-russian-military-uses-its-drones-ukraine-2022-03-28/>) from the original on 28 March 2022. Retrieved 28 March 2022.
104. "Chinese drone giant DJI suspends business in Russia, Ukraine" (<https://www.aljazeera.com/economy/2022/4/27/chinese-drone-giant-dji-suspends-business-in-russia-ukraine>). *Al Jazeera*. 27 April 2022. Archived (<https://web.archive.org/web/20220615225846/https://www.aljazeera.com/economy/2022/4/27/chinese-drone-giant-dji-suspends-business-in-russia-ukraine>) from the original on 15 June 2022. Retrieved 27 April 2022.
105. Kirton, David (27 April 2022). "China's DJI halts Russia, Ukraine sales to prevent use of its drones in combat" (<https://www.reuters.com/technology/chinese-drone-maker-dji-suspends-business-activities-russia-ukraine-2022-04-26/>). *Reuters*. Archived (<https://web.archive.org/web/20220426235354/https://www.reuters.com/technology/chinese-drone-maker-dji-suspends-business-activities-russia-ukraine-2022-04-26/>) from the original on 26 April 2022. Retrieved 27 April 2022.
106. Sabbagh, Dan; Mazhulin, Artem (10 April 2023). "'They're starting to die': fears Ukraine's drone supremacy may soon be over" (<https://www.theguardian.com/world/2023/apr/10/in-war-there-are-no-emotions-ukraine-drone-squads-bakhmut>). *The Guardian*. Archived (<https://web.archive.org/web/20240815030948/https://www.theguardian.com/world/2023/apr/10/in-war-there-are-no-emotions-ukraine-drone-squads-bakhmut>) from the original on 15 August 2024. Retrieved 24 July 2024.
107. Standish, Reid (5 October 2023). "Chinese Drones Flow To Training Centers Linked To Russian War In Ukraine" (<https://www.rferl.org/a/russia-ukraine-chinese-drones-training-centers/32621432.html>). *Radio Free Europe*. Archived (<https://web.archive.org/web/20240815020956/https://www.rferl.org/a/russia-ukraine-chinese-drones-training-centers/32621432.html>) from the original on 15 August 2024. Retrieved 24 July 2024.
108. Mozur, Paul; Hopkins, Valerie (30 September 2023). "Ukraine's War of Drones Runs Into an Obstacle: China" (<https://www.nytimes.com/2023/09/30/technology/ukraine-russia-war-drones-china.html>). *The New York Times*. Archived (<https://web.archive.org/web/20240803132222/https://www.nytimes.com/2023/09/30/technology/ukraine-russia-war-drones-china.html>) from the original on 3 August 2024. Retrieved 24 July 2024.
109. "Design, Simulation and New Applications of Unmanned Aerial Vehicles" (<https://www.mdpi.com/topics/UAV>). *www.mdpi.com*. Retrieved 24 March 2023.
110. Nagel, Huub; Bondt, Geert; Custers, Bart; Vergouw, Bas (16 July 2016). "Drone Technology: Types, Payloads, Applications, Frequency Spectrum Issues and Future Developments" (<https://www.researchgate.net/publication/309184029>). *The Future of Drone Use*.

111. da Silva, F.B; Scott, S.D; Cummings, M.L (December 2007). "Design Methodology for Unmanned Aerial Vehicle (UAV) Team Coordination" (https://dspace.mit.edu/bitstream/handle/1721.1/46732/HAL2007_05.pdf) (PDF). *Design Methodology for Unmanned Aerial Vehicle (UAV) Team Coordination*.
112. Torres-Sánchez, Jorge; López-Granados, Francisca; De Castro, Ana Isabel; Peña-Barragán, José Manuel (6 March 2013). "Configuration and Specifications of an Unmanned Aerial Vehicle (UAV) for Early Site Specific Weed Management" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3590160>). *PLOS ONE*. **8** (3) e58210. Bibcode:2013PLoSO...858210T (<https://ui.adsabs.harvard.edu/abs/2013PLoSO...858210T>). doi:10.1371/journal.pone.0058210 (<https://doi.org/10.1371%2Fjournal.pone.0058210>). PMC 3590160 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3590160>). PMID 23483997 (<https://pubmed.ncbi.nlm.nih.gov/23483997>).
113. "Model airplane history-maker Maynard Hill dies at the age of 85" (https://www.washingtonpost.com/local/obituaries/model-airplane-history-maker-maynard-hill-dies-at-the-age-of-85/2011/06/08/AGcnyQNH_story.html). *The Washington Post*. Archived (https://web.archive.org/web/20180704034821/https://www.washingtonpost.com/local/obituaries/model-airplane-history-maker-maynard-hill-dies-at-the-age-of-85/2011/06/08/AGcnyQNH_story.html) from the original on 4 July 2018. Retrieved 17 May 2018.
114. Trakimavicius, Lukas (15 October 2025). "Wanted: More Batteries for Defence" (<https://www.rusi.org/explore-our-research/publications/commentary/wanted-more-batteries-defence>). *RUSI*.
115. Savvaris, Al; Xie, Ye; Malandrakis, Konstantinos; Lopez, Matias; Tsourdos, Antonios (2016). "Development of a fuel cell hybrid-powered unmanned aerial vehicle". *2016 24th Mediterranean Conference on Control and Automation (MED)*. pp. 1242–1247. doi:10.1109/MED.2016.7536038 (<https://doi.org/10.1109%2FMED.2016.7536038>). hdl:1826/14084 (<https://hdl.handle.net/1826%2F14084>). ISBN 978-1-4673-8345-5.
116. Baldic, Jeff; Osenar, Paul; Lauder, Nick; Launie, Peter (2010). "Fuel cell systems for long duration electric UAVs and UGVs". In Suresh, Raja (ed.). *Defense Transformation and Net-Centric Systems 2010*. Vol. 7707. doi:10.1117/12.851779 (<https://doi.org/10.1117%2F12.851779>).
117. Chu, Deryn; Jiang, R.; Dunbar, Z.; Grew, Kyle; McClure, J. (2015). "Fuel cell powered small unmanned aerial systems (UASs) for extended endurance flights". In Karlsen, Robert E.; Gage, Douglas W.; Shoemaker, Charles M.; Gerhart, Grant R. (eds.). *Unmanned Systems Technology XVII*. Vol. 9468. pp. 94680E. doi:10.1117/12.2087336 (<https://doi.org/10.1117%2F12.2087336>).
118. Kajal, Kapil. "China's invisible winged drones fly like birds to hit targets silently" (<https://interestingengineering.com/military/china-invisible-drones-flap-wings-like-birds>). *Interesting Engineering*. Retrieved 30 August 2026.
119. Chirarattananon, Pakpong; Ma, Kevin Y.; Wood, Robert J. (2014). "Adaptive control of a millimeter-scale flapping-wing robot". *Bioinspiration & Biomimetics*. **9** (2) 025004. Bibcode:2014BiBi....9b5004C (<https://ui.adsabs.harvard.edu/abs/2014BiBi....9b5004C>). doi:10.1088/1748-3182/9/2/025004 (<https://doi.org/10.1088%2F1748-3182%2F9%2F2%2F025004>). PMID 24855052 (<https://pubmed.ncbi.nlm.nih.gov/24855052>).
120. Sarah Knapton (29 March 2016). "Giant remote-controlled beetles and 'biobot' insects could replace drones" (<https://web.archive.org/web/20160401144132/http://www.telegraph.co.uk/science/2016/03/29/giant-remote-controlled-beetles-could-replace-drones/>). *The Telegraph*. Archived from the original (<https://www.telegraph.co.uk/science/2016/03/29/giant-remote-controlled-beetles-could-replace-drones/>) on 1 April 2016.
121. Antonio (11 July 2024). "EASA Approves ETSO Certification Basis for Veronte Autopilot" (<https://www.embention.com/en/news/easa-approves-etso-certification-basis-for-veronte-autopilot/>). *Embention*. Retrieved 2 August 2024.

122. Floreano, Dario; Wood, Robert J. (27 May 2015). "Science, technology and the future of small autonomous drones" (<http://infoscience.epfl.ch/record/208757>). *Nature*. **521** (7553): 460–466. Bibcode:2015Natur.521..460F (<https://ui.adsabs.harvard.edu/abs/2015Natur.521..460F>). doi:10.1038/nature14542 (<https://doi.org/10.1038%2Fnature14542>). PMID 26017445 (<https://pubmed.ncbi.nlm.nih.gov/26017445>).
123. Fasano, Giancarmine; Accardo, Domenico; Tirri, Anna Elena; Moccia, Antonio; De Lellis, Ettore (1 October 2015). "Radar/electro-optical data fusion for non-cooperative UAS sense and avoid" (<https://doi.org/10.1016%2Fj.ast.2015.08.010>). *Aerospace Science and Technology*. **46**: 436–450. Bibcode:2015AeST...46..436F (<https://ui.adsabs.harvard.edu/abs/2015AeST...46..436F>). doi:10.1016/j.ast.2015.08.010 (<https://doi.org/10.1016%2Fj.ast.2015.08.010>).
124. "Arduino Playground – WhatIsDegreesOfFreedom6DOF9DOF10DOF11DOF" (<https://playground.arduino.cc/Main/WhatIsDegreesOfFreedom6DOF9DOF10DOF11DOF>). *playground.arduino.cc*. Archived (<https://web.archive.org/web/20160218110500/http://playground.arduino.cc/Main/WhatIsDegreesOfFreedom6DOF9DOF10DOF11DOF>) from the original on 18 February 2016. Retrieved 4 February 2016.
125. Manfreda, Salvatore; McCabe, Matthew F.; Miller, Pauline E.; Lucas, Richard; Pajuelo Madrigal, Victor; Mallinis, Giorgos; Ben Dor, Eyal; Helman, David; Estes, Lyndon; Ciraolo, Giuseppe; Müllerová, Jana; Tauro, Flavia; De Lima, M. Isabel; De Lima, João L. M. P.; Maltese, Antonino; Frances, Felix; Caylor, Kelly; Kohv, Marko; Perks, Matthew; Ruiz-Pérez, Guiomar; Su, Zhongbo; Vico, Giulia; Toth, Brigitta (20 April 2018). "On the Use of Unmanned Aerial Systems for Environmental Monitoring" (<https://doi.org/10.3390%2Frs10040641>). *Remote Sensing*. **10** (4): 641. Bibcode:2018RemS...10..641M (<https://ui.adsabs.harvard.edu/abs/2018RemS...10..641M>). doi:10.3390/rs10040641 (<https://doi.org/10.3390%2Frs10040641>). hdl:10251/127481 (<https://hdl.handle.net/10251%2F127481>).
126. "7.3. Sensing for Drones—Introduction to Robotics and Perception" (https://www.roboticsbook.org/S73_drone_sensing.html). *www.roboticsbook.org*.
127. "7.5. Trajectory Optimization—Introduction to Robotics and Perception" (https://www.roboticsbook.org/S75_drone_planning.html). *www.roboticsbook.org*.
128. Deshpande, Aditya M.; Minai, Ali A.; Kumar, Manish (2021). "Robust Deep Reinforcement Learning for Quadcopter Control". *IFAC-PapersOnLine*. **54** (20): 90–95. doi:10.1016/j.ifacol.2021.11.158 (<https://doi.org/10.1016%2Fj.ifacol.2021.11.158>).
129. Manuel, Rojoef (20 November 2024). "US Navy's 14th Freedom Littoral Combat Ship Enters Service" (<https://thedefensepost.com/2024/11/20/us-14th-freedom-enters-service/>). *The Defense Post*.
130. Pierre-Jean Bristeau; François Callou; David Vissière; Nicolas Petit (2011). "The Navigation and Control technology inside the AR.Drone micro UAV" (<http://www.nt.ntnu.no/users/skoge/prost/proceedings/ifac11-proceedings/data/html/papers/2327.pdf>) (PDF). *IFAC World Congress*. Archived (<https://web.archive.org/web/20230227213841/https://folk.ntnu.no/skoge/prost/proceedings/ifac11-proceedings/data/html/papers/2327.pdf>) (PDF) from the original on 27 February 2023. Retrieved 4 February 2016.
131. "Cellular enables safer drone deployments" (<https://www.qualcomm.com/invention/technologies/lte/advanced-pro/cellular-drone-communication>). *Qualcomm*. Archived (<https://web.archive.org/web/20180509151213/https://www.qualcomm.com/invention/technologies/lte/advanced-pro/cellular-drone-communication>) from the original on 9 May 2018. Retrieved 9 May 2018.
132. "Identifying Critical Manned-Unmanned Teaming Skills for Unmanned Aircraft System Operators" (<http://apps.dtic.mil/dtic/tr/fulltext/u2/a565510.pdf>) (PDF). *U.S. Army Research Institute for the Behavioral and Social Sciences*. September 2012. Archived (<https://web.archive.org/web/20160206104148/http://www.dtic.mil/dtic/tr/fulltext/u2/a565510.pdf>) (PDF) from the original on 6 February 2016.

133. "Minimum requirements related to technical performance for IMT-2020 radio interface(s)" (<https://www.itu.int/pub/R-REP-M.2410-2017>). *www.itu.int*. Archived (<https://web.archive.org/web/20200806202716/https://www.itu.int/pub/R-REP-M.2410-2017>) from the original on 6 August 2020. Retrieved 8 October 2020.
134. Vinogradov, Evgenii; Kumar, A. V. S. Sai Bhargav; Minucci, Franco; Pollin, Sofie; Natalizio, Enrico (2023). "Remote ID for separation provision and multi-agent navigation". 2023 *IEEE/AIAA 42nd Digital Avionics Systems Conference (DASC)*. pp. 1–10. arXiv:2309.00843 (<https://arxiv.org/abs/2309.00843>). doi:10.1109/DASC58513.2023.10311133 (<https://doi.org/10.1109%2FDASC58513.2023.10311133>). ISBN 979-8-3503-3357-2.
135. "Automated Vehicles for Safety" (<https://www.nhtsa.gov/technology-innovation/automated-vehicles-safety>). United States: National Highway Traffic Safety Administration. Archived (<https://web.archive.org/web/20211007021013/https://www.nhtsa.gov/technology-innovation/automated-vehicles-safety>) from the original on 7 October 2021. Retrieved 8 October 2021.
136. Clough, Bruce (August 2002). "Metrics, Schmetrics! How The Heck Do You Determine A UAV's Autonomy Anyway?" (<https://apps.dtic.mil/sti/citations/ADA515926>). *US Air Force Research Laboratory*. Archived (<https://web.archive.org/web/20200924155100/https://apps.dtic.mil/sti/citations/ADA515926>) from the original on 24 September 2020.
137. Davenport, Christian (23 April 2015). "Watch a step in Navy history: an autonomous drone gets refueled mid-air" (<https://www.washingtonpost.com/news/checkpoint/wp/2015/04/23/watch-a-step-in-aviation-history-an-autonomous-drone-getting-refueled-mid-air/>). *The Washington Post*. Archived (<https://web.archive.org/web/20160120103200/https://www.washingtonpost.com/news/checkpoint/wp/2015/04/23/watch-a-step-in-aviation-history-an-autonomous-drone-getting-refueled-mid-air/>) from the original on 20 January 2016. Retrieved 3 February 2016.
138. Tahir, Anam; Böling, Jari; Haghbayan, Mohammad-Hashem; Toivonen, Hannu T.; Plosila, Juha (2019). "Swarms of Unmanned Aerial Vehicles – A Survey" (<https://doi.org/10.1016%2Fj.jii.2019.100106>). *Journal of Industrial Information Integration*. **16** 100106. doi:10.1016/j.jii.2019.100106 (<https://doi.org/10.1016%2Fj.jii.2019.100106>).
139. "Teaching tiny drones how to fly themselves" (<https://arstechnica.com/information-technology/2012/11/a-beautiful-robotic-mind/1/>). *Ars Technica*. 27 November 2012. Archived (<https://web.archive.org/web/20160205171813/http://arstechnica.com/information-technology/2012/11/a-beautiful-robotic-mind/1/>) from the original on 5 February 2016. Retrieved 4 February 2016.
140. "Biomimetics and Dextrous Manipulation Lab – MultiModalRobots" (<http://bdml.stanford.edu/Main/MultiModalRobots>). *bdml.stanford.edu*. Archived (<https://web.archive.org/web/20160323135154/http://bdml.stanford.edu/Main/MultiModalRobots>) from the original on 23 March 2016. Retrieved 21 March 2016.
141. D'Andrea, Raffaello (11 June 2013). "The astounding athletic power of quadcopters" (https://www.ted.com/talks/raffaello_d_andrea_the_astounding_athletic_power_of_quadcopters#t-1461). *www.ted.com*. Archived (https://web.archive.org/web/20160205181354/http://www.ted.com/talks/raffaello_d_andrea_the_astounding_athletic_power_of_quadcopters#t-1461) from the original on 5 February 2016. Retrieved 4 February 2016.
142. Yanguo, Song; Huanjin, Wang (1 June 2009). "Design of Flight Control System for a Small Unmanned Tilt Rotor Aircraft" (<https://doi.org/10.1016%2FS1000-9361%2808%2960095-3>). *Chinese Journal of Aeronautics*. **22** (3): 250–256. Bibcode:2009ChJAn..22..250Y (<https://ui.adsabs.harvard.edu/abs/2009ChJAn..22..250Y>). doi:10.1016/S1000-9361(08)60095-3 (<https://doi.org/10.1016%2FS1000-9361%2808%2960095-3>).

143. "The device, designed for landing UAV helicopter type on a flat vertical surface" (<https://patents.google.com/patent/RU160508U1/en?inventor=%D0%90%D0%BD%D0%B4%D1%80%D0%B5%D0%B9+%D0%98%D0%B2%D0%B0%D0%BD%D0%BE%D0%B2%D0%B8%D1%87+%D0%91%D0%BE%D0%B4%D1%80%D0%B5%D0%BD%D0%BA%D0%BE>). *patents.google.com*. Archived (<https://web.archive.org/web/20170307103055/https://patents.google.com/patent/RU160508U1/en?inventor=%D0%90%D0%BD%D0%B4%D1%80%D0%B5%D0%B9+%D0%98%D0%B2%D0%B0%D0%BD%D0%BE%D0%B2%D0%B8%D1%87+%D0%91%D0%BE%D0%B4%D1%80%D0%B5%D0%BD%D0%BA%D0%BE>) from the original on 7 March 2017. Retrieved 6 November 2016.
144. "The Importance of Proper Cooling and Airflow for Optimal Drone Performance" (<https://www.pelonistechnologies.com/blog/proper-cooling-drone-performance>). *Pelonis Technologies*. Archived (<https://web.archive.org/web/20180622220335/https://www.pelonistechnologies.com/blog/proper-cooling-drone-performance>) from the original on 22 June 2018. Retrieved 22 June 2018.
145. "Flying on Hydrogen: Georgia Tech Researchers Use Fuel Cells to Power Unmanned Aerial Vehicle | Georgia Tech Research Institute" (<http://www.gtri.gatech.edu/casestudy/flying-hydrogen>). *www.gtri.gatech.edu*. Archived (<https://web.archive.org/web/20160203120006/http://www.gtri.gatech.edu/casestudy/flying-hydrogen>) from the original on 3 February 2016. Retrieved 4 February 2016.
146. "Hydrogen-powered Hycopter quadcopter could fly for 4 hours at a time" (<http://www.gizmag.com/horizon-energy-systems-hycopter-fuel-cell-drone/37585/>). *www.gizmag.com*. 20 May 2015. Archived (<https://web.archive.org/web/20160204194507/http://www.gizmag.com/horizon-energy-systems-hycopter-fuel-cell-drone/37585/>) from the original on 4 February 2016. Retrieved 4 February 2016.
147. Gibbs, Yvonne (31 March 2015). "NASA Armstrong Fact Sheet: Beamed Laser Power for UAVs" (<https://www.nasa.gov/centers/armstrong/news/FactSheets/FS-087-DFRC.html>). NASA. Archived (<https://web.archive.org/web/20190405021645/https://www.nasa.gov/centers/armstrong/news/FactSheets/FS-087-DFRC.html>) from the original on 5 April 2019. Retrieved 22 June 2018.
148. *Vertical Challenge: "Monsters of the sky"* (<https://web.archive.org/web/20130911145056/http://www.hiller.org/files/docs/2003Q3.pdf>) (PDF), archived from the original (<http://www.hiller.org/files/docs/2003Q3.pdf>) (PDF) on 11 September 2013
149. "General Atomics Gnat" (<http://www.designation-systems.net/dusrm/app4/gnat.html>). Designation-systems.net. Archived (<https://web.archive.org/web/20081211063639/http://www.designation-systems.net/dusrm/app4/gnat.html>) from the original on 11 December 2008. Retrieved 8 January 2015.
150. "UAV Notes" (https://web.archive.org/web/20130730032004/http://www.vectorsite.net/twuav_12.html). Archived from the original on 30 July 2013.
151. "Trans atlantic Model" (<http://arquivo.pt/wayback/20160522211703/http://tam.plannet21.com/>). Tam.plannet21.com. Archived from the original (<http://tam.plannet21.com/>) on 22 May 2016. Retrieved 8 January 2015.
152. "QinetiQ's Zephyr UAV exceeds official world record for longest duration unmanned flight" (https://web.archive.org/web/20110423162325/http://www.qinetiq.com/home/newsroom/news_releases_homepage/2007/3rd_quarter/qinetiq_s_zephyr_uav.html) (Press release). QinetiQ. 10 September 2007. Archived from the original (http://www.qinetiq.com/home/newsroom/news_releases_homepage/2007/3rd_quarter/qinetiq_s_zephyr_uav.html) on 23 April 2011.
153. Simonite, Tom. "New Scientist Technology Blog: Solar plane en route to everlasting flight" (<https://web.archive.org/web/20150402090342/http://www.newscientist.com/blog/technology/2007/09/solar-flyer-en-route-to-everlasting.html>). *New Scientist*. Archived from the original (<https://web.archive.org/web/20150402090342/http://www.newscientist.com/blog/technology/2007/09/solar-flyer-en-route-to-everlasting.html>) on 2 April 2015. Retrieved 8 January 2015.

154. "Northrop Grumman's Global Hawk Unmanned Aircraft Sets 33-Hour Flight Endurance Record" (http://www.spacewar.com/reports/Northrop_Grumman_Global_Hawk_Unmanned_Aircraft_Sets_33_Hour_Flight_Endurance_Record_999.html). Spacewar.com. Retrieved 27 August 2013.
155. "QinetiQ's Zephyr UAV flies for three and a half days to set unofficial world record for longest duration unmanned flight" (https://web.archive.org/web/20110524020631/http://www.qinetiq.com/home/newsroom/news_releases_homepage/2008/3rd_quarter/qinetiq_s_zephyr_uav.html) (Press release). QinetiQ. 24 August 2008. Archived from the original (http://www.qinetiq.com/home/newsroom/news_releases_homepage/2008/3rd_quarter/qinetiq_s_zephyr_uav.html) on 24 May 2011.
156. "QinetiQ files for three world records for its Zephyr Solar powered UAV" (https://web.archive.org/web/20100924231501/http://www.qinetiq.com/home/newsroom/news_releases_homepage/2010/3rd_quarter/qinetiq_files_for.html). QinetiQ (Press release). 24 August 2010. Archived from the original (http://www.qinetiq.com/home/newsroom/news_releases_homepage/2010/3rd_quarter/qinetiq_files_for.html) on 24 September 2010.
157. MacDonald, Alistair (14 July 2023). "Drones Reach Stratospheric Heights in Race to Fly Higher, Longer" (<https://www.wsj.com/articles/drones-reach-stratospheric-heights-in-race-to-fly-higher-longer-a29ceea3>). *The Wall Street Journal* – via www.wsj.com.
158. Boniol (December 2014). "Towards Modular and Certified Avionics for UAV" (http://www.aerospacelab-journal.org/sites/www.aerospacelab-journal.org/files/AL08-02_0.pdf) (PDF). *Aerospacelab Journal*. Archived (https://web.archive.org/web/20160204200527/http://www.aerospacelab-journal.org/sites/www.aerospacelab-journal.org/files/AL08-02_0.pdf) (PDF) from the original on 4 February 2016. Retrieved 4 February 2016.
159. D. Boskovic and Knoebel (2009). "A Comparison Study of Several Adaptive Control Strategies for Resilient Flight Control" (<https://web.archive.org/web/20160204191326/http://enu.kz/repository/2009/AIAA-2009-5736.pdf>) (PDF). *AIAA Guidance, Navigation and Control Conference*. Archived from the original (<http://enu.kz/repository/2009/AIAA-2009-5736.pdf>) (PDF) on 4 February 2016.
160. Atkins. "Certifiable Autonomous Flight Management for Unmanned Aircraft Systems" (<http://www.naefrontiers.org/File.aspx?id=25848>). *University of Michigan*. Archived (<https://web.archive.org/web/20170305144615/https://www.naefrontiers.org/File.aspx?id=25848>) from the original on 5 March 2017. Retrieved 4 February 2016.
161. Subhav Pradhan; William Otte; Abhishek Dubey; Aniruddha Gokhale; Gabor Karsai (2013). "Key Considerations for a Resilient and Autonomous Deployment and Configuration Infrastructure for Cyber-Physical Systems" (http://www.dre.vanderbilt.edu/~gokhale/WWW/papers/EASe14_AutonomousDnC.pdf) (PDF). *Dept. of Electrical Engineering and Computer Science Vanderbilt University, Nashville*. Archived (https://web.archive.org/web/20160204194114/http://www.dre.vanderbilt.edu/~gokhale/WWW/papers/EASe14_AutonomousDnC.pdf) (PDF) from the original on 4 February 2016. Retrieved 4 February 2016.
162. "How Autonomous Drone Flights Will Go Beyond Line of Sight" (<https://www.nanalyze.com/2019/12/autonomous-drone-flights/>). *Nanalyze*. 31 December 2019. Archived (<https://web.archive.org/web/20200520194214/https://www.nanalyze.com/2019/12/autonomous-drone-flights/>) from the original on 20 May 2020. Retrieved 16 April 2020.
163. McNabb, Miriam (28 February 2020). "Drones Get the Lights Back on Faster for Florida Communities" (<https://dronelife.com/2020/02/28/drones-get-the-lights-back-on-faster-for-florida-communities/>). *DRONELIFE*. Archived (<https://web.archive.org/web/20200312063842/https://dronelife.com/2020/02/28/drones-get-the-lights-back-on-faster-for-florida-communities/>) from the original on 12 March 2020. Retrieved 16 April 2020.

164. Peck, Abe (19 March 2020). "Coronavirus Spurs Percepto's Drone-in-a-Box Surveillance Solution" (<https://insideunmannedsystems.com/unintended-consequences-coronavirus-spurs-perceptos-drone-in-a-box-surveillance-solution/>). *Inside Unmanned Systems*. Archived (<https://web.archive.org/web/20200324195138/https://insideunmannedsystems.com/unintended-consequences-coronavirus-spurs-perceptos-drone-in-a-box-surveillance-solution/>) from the original on 24 March 2020. Retrieved 16 April 2020.
165. Valle, Roberto G. (January 2022). "Rapid drone semi-automated counts of wintering Greater Flamingos (*Phoenicopterus roseus*) as a tool for amateur researchers". *Ibis*. **164** (1): 320–328. doi:10.1111/ibi.12993 (<https://doi.org/10.1111%2Fibi.12993>).
166. Mademlis, Ioannis; Nikolaidis, Nikos; Tefas, Anastasios; Pitas, Ioannis; Wagner, Tilman; Messina, Alberto (30 September 2020). "Autonomous UAV Cinematography: A Tutorial and a Formalized Shot-Type Taxonomy". *ACM Computing Surveys*. **52** (5): 1–33. doi:10.1145/3347713 (<https://doi.org/10.1145%2F3347713>).
167. Johnson, Lizzie; Korolchuk, Serhii; Galouchka, Anastacia; Khudov, Kostiantyn; Ram, Ed; Chen, Yutao; Ledur, Júlia; Sand, Bishop (8 September 2025). "Russia bombs Ukraine almost every night. This is what it sounds like" (<https://www.washingtonpost.com/world/interactive/2025/ukraine-bombing-sounds-war-sirens-russia/>). *The Washington Post*. Archived (<https://web.archive.org/web/20250909004516/https://www.washingtonpost.com/world/interactive/2025/ukraine-bombing-sounds-war-sirens-russia/>) from the original on 9 September 2025.
168. "Ukraine's Hornet UAVs Targeting Crimea Corridor Could Fly Farther, Longer With Balloon Deployment" (https://en.defence-ua.com/weapon_and_tech/ukraines_hornet_uavs_targeting_crimea_corridor_could_fly_farther_longer_with_balloon_deployment-18586.html). *Defense Express*. 24 May 2026. Retrieved 24 May 2026.
169. Horowitz, Michael C. (2020). "Do Emerging Military Technologies Matter for International Politics?" (<https://doi.org/10.1146%2Fannurev-polisci-050718-032725>). *Annual Review of Political Science*. **23** (1): 385–400. doi:10.1146/annurev-polisci-050718-032725 (<https://doi.org/10.1146%2Fannurev-polisci-050718-032725>).
170. "BAYKAR, THE GLOBAL LEADER IN UCACV EXPORTS, ACHIEVES \$1.8 BILLION IN EXPORTS IN 2024" (<https://baykartech.com/en/press/baykar-the-global-leader-in-ucav-exports-achieves-18-billion-in-exports-in-2024/>). Baykar Tech. 4 February 2025. Archived (<https://web.archive.org/web/20250210075111/https://baykartech.com/en/press/baykar-the-global-leader-in-ucav-exports-achieves-18-billion-in-exports-in-2024/>) from the original on 10 February 2025. Retrieved 10 February 2025.
171. "Strengthening Turkish Policy on Drone Exports" (<https://carnegieendowment.org/research/2022/01/strengthening-turkish-policy-on-drone-exports?lang=en>). Carnegie Endowment for International. 18 January 2022. Archived (<https://web.archive.org/web/20220323025634/https://carnegieendowment.org/2022/01/18/strengthening-turkish-policy-on-drone-exports-pub-86183>) from the original on 23 March 2022. Retrieved 17 March 2022.
172. "Turkey's defense industry targets more than \$4 billion in exports this year: Official" (<https://www.hurriyetdailynews.com/turkeys-defense-industry-targets-more-than-4-billion-in-exports-this-year-official-172005>). *Hürriyet Daily News*. 6 March 2022. Archived (<https://web.archive.org/web/20220317015136/https://www.hurriyetdailynews.com/turkeys-defense-industry-targets-more-than-4-billion-in-exports-this-year-official-172005>) from the original on 17 March 2022. Retrieved 17 March 2022.
173. "Combat drones in China are coming to a conflict near you" (<https://www.intelligent-aerospace.com/unmanned/article/14199662/china-military-drones-uav-unmanned>). *www.intelligent-aerospace.com*. 19 March 2021. Archived (<https://web.archive.org/web/20210607015056/https://www.intelligent-aerospace.com/unmanned/article/14199662/china-military-drones-uav-unmanned>) from the original on 7 June 2021. Retrieved 7 June 2021.

174. "Market for Military Drones will Surge" (<https://insideunmannedsystems.com/market-military-drones-will-surge/>). 27 October 2016. Archived (<https://web.archive.org/web/20180219152237/http://insideunmannedsystems.com/market-military-drones-will-surge/>) from the original on 19 February 2018. Retrieved 19 February 2018.
175. "Turkish defence industry grows as Akinci UCAV signs first export deal" (<https://www.trtworld.com/turkey/turkish-defence-industry-grows-as-akinci-ucav-signs-first-export-deal-53987>). TRTWORLD. 23 January 2022. Archived (<https://web.archive.org/web/20220130161059/http://www.trtworld.com/turkey/turkish-defence-industry-grows-as-akinci-ucav-signs-first-export-deal-53987>) from the original on 30 January 2022. Retrieved 17 March 2022.
176. Arnett, George (16 March 2015). "The numbers behind the worldwide trade in UAVs" (<http://www.theguardian.com/news/datablog/2015/mar/16/numbers-behind-worldwide-trade-in-drones-uk-israel>). *The Guardian*. Archived (<https://web.archive.org/web/20161219153019/http://www.theguardian.com/news/datablog/2015/mar/16/numbers-behind-worldwide-trade-in-drones-uk-israel>) from the original on 19 December 2016. Retrieved 13 December 2016.
177. "Pentagon Plans for Cuts to Drone Budgets" (<https://web.archive.org/web/20150108232952/http://www.dodbuzz.com/2014/01/02/pentagon-plans-for-cuts-to-drone-budgets/>). *DoD Buzz*. 2 January 2014. Archived from the original (<http://www.dodbuzz.com/2014/01/02/pentagon-plans-for-cuts-to-drone-budgets/>) on 8 January 2015. Retrieved 17 March 2022.
178. "Is China at the Forefront of Drone Technology?" (<https://chinapower.csis.org/china-drones-unmanned-technology/>). *Center for Strategic and International Studies*. 29 May 2018.
179. Seong Hyeon Choi (25 November 2023). "Drone tech gives China an edge in Middle East arms sales, but Israel-Gaza war brings risks: analysts" (<https://www.scmp.com/news/china/military/article/3242630/drone-tech-gives-china-edge-middle-east-arms-sales-israel-gaza-war-brings-risks-analysts>). *South China Morning Post*.
180. Somerville, Heather (9 November 2023). "Chinese, American—It Doesn't Matter. Israel Wants Inexpensive Drones" (<https://www.wsj.com/politics/national-security/chinese-american-it-doesnt-matter-israel-wants-inexpensive-drones-339a1355?mod=mhp>). *The Wall Street Journal*.
181. Skove, Sam (1 May 2024). "UK eyes Chinese drone parts for Ukraine" (<https://www.defenseone.com/technology/2024/04/uk-eyes-chinese-drone-parts-ukraine/396211/>). *Defense One*.
182. Robbert van Rossum (17 November 2025). "De UAS-dreiging in de Russisch–Oekraïense oorlog" (<https://militairespectator.nl/artikelen/de-uas-dreiging-de-russisch-oekraïense-oorlog>). *Militaire Spectator* (in Dutch).
183. Lister, Tim; Gak, Kosta (22 November 2025). "Russia's drone revolution heaps pressure on Ukrainian defenses" (<https://edition.cnn.com/2025/11/22/europe/russia-rubicon-unit-drone-revolution-ukraine-intl-cmd>). CNN. Retrieved 13 December 2025.
184. Cherevko, Vlad. "Drone from Rubizh Brigade delivers bike to save lone wounded soldier on Siversk front – video" (<https://www.pravda.com.ua/eng/news/2025/07/31/7524193/>). *Ukrainska Pravda*. Retrieved 23 December 2025.
185. "«Волокуші» з неба: засідка, щільний вогонь і FPV-дрони — як бійці 225-го полку врятували побратима" (<https://armyinform.com.ua/2025/12/11/volokushi-z-neba-zasidka-shilnyj-vogon-i-fpv-drony-yak-bijci-225-go-polku-vryatuvaty-pobratyma/>) (in Ukrainian). 11 December 2025. Retrieved 23 December 2025.
186. "Ukrainian copy of Russian Molniya drone already in use by 60 units – video" (<https://www.pravda.com.ua/eng/news/2026/01/28/8018310/>). *Ukrainska Pravda*. Retrieved 6 April 2026.
187. Axe, David (5 June 2025). "Kyiv, we have a problem: Russia just reverse-engineered Ukraine's drone motherships" (<https://euromaidanpress.com/2025/06/05/molniya-drone-mothership/>). *Euromaidan Press*. Retrieved 6 April 2026.
188. "Cost asymmetry in Gulf air defense exposes long-term sustainability risks" (<https://www.aa.com.tr/en/middle-east/cost-asymmetry-in-gulf-air-defense-exposes-long-term-sustainability-risks/3847942>). *Anadolu* (in Turkish). Retrieved 6 April 2026.

189. Srinivasan, Chandrashekar (10 March 2026). "With Missiles Carrying 1,000 kg Explosives, A New Tactical Phase For Iran" (<https://www.ndtv.com/world-news/iran-israel-war-iran-1000kg-1-ton-missile-warheads-iran-fires-khorramshahr-kheibar-shekan-missiles-11193972>). *NDTV*. Retrieved 6 April 2026.
190. Chatterjee, Nandika; Hall, Richard (15 March 2026). "The Iran War Created Demand for Cheap Drone Interceptors. Now Zelensky Holds the Cards" (<https://time.com/article/2026/03/15/iran-war-ukraine-drones/>). Retrieved 6 April 2026.
191. "New Report Unveils Drone Industry Market Share Figures" (<https://dronelife.com/2018/09/18/new-report-unveils-drone-industry-market-share-figures/>). *Drone Life*. 18 September 2018.
192. "DJI market share: here's exactly how rapidly it has grown in just a few years" (<http://thedronegirl.com/2018/09/18/dji-market-share/>). *Emberify Blog*. 18 September 2018. Archived (<http://web.archive.org/web/20180924033312/http://thedronegirl.com/2018/09/18/dji-market-share/>) from the original on 24 September 2018. Retrieved 18 September 2018.
193. Anwar, Nessa (7 February 2023). "World's largest drone maker is unfazed—even if it is blacklisted by the U.S." (<https://www.cnbc.com/2023/02/08/worlds-largest-drone-maker-dji-is-unfazed-by-challenges-like-us-blacklist.html>) *CNBC*.
194. Friedman, Lisa; McCabe, David (29 January 2020). "Interior Dept. Grounds Its Drones Over Chinese Spying Fears" (<https://www.nytimes.com/2020/01/29/technology/interior-chinese-drones.html>). *The New York Times*. Archived (<https://web.archive.org/web/20200129191125/https://www.nytimes.com/2020/01/29/technology/interior-chinese-drones.html>) from the original on 29 January 2020. Retrieved 17 November 2020.
195. Miller, Maggie (8 October 2020). "DOJ bans use of grant funds for certain foreign-made drones" (<https://thehill.com/policy/cybersecurity/520269-justice-department-issues-policy-banning-use-of-grant-funds-for-certain>). *The Hill*. Archived (<https://web.archive.org/web/20201128001142/https://thehill.com/policy/cybersecurity/520269-justice-department-issues-policy-banning-use-of-grant-funds-for-certain>) from the original on 28 November 2020. Retrieved 17 November 2020.
196. Daly, David (2021). "5 Major Drone Manufacturers Powering the Industry" (<https://consortiq.com/uas-resources/5-major-drone-manufacturers-powering-the-industry>). *Consortiq*.
197. Oliver, Matt (15 May 2025). "China dominates the drone industry. The West needs to change that fast" (<https://www.telegraph.co.uk/business/2025/05/15/china-dominates-the-drone-industry-the-west-needs-to-change/>). *The Telegraph*.
198. Pusztaszeri, Aosheng (16 December 2024). "Why China's UAV Supply Chain Restrictions Weaken Ukraine's Negotiating Power" (<https://www.csis.org/analysis/why-chinas-uav-supply-chain-restrictions-weaken-ukraines-negotiating-power>). *Center for Strategic and International Studies*.
199. "UAS by the Numbers" (https://www.faa.gov/uas/resources/by_the_numbers/). *www.faa.gov*. Archived (https://web.archive.org/web/20210517144023/https://www.faa.gov/uas/resources/by_the_numbers/) from the original on 17 May 2021. Retrieved 24 May 2021.
200. "Consumer Drones By the Numbers in 2018 and Beyond | News Ledge" (<https://www.newsledge.com/consumer-drones-2018-numbers/>). *News Ledge*. 4 April 2017. Archived (<https://web.archive.org/web/20181014010203/https://www.newsledge.com/consumer-drones-2018-numbers/>) from the original on 14 October 2018. Retrieved 13 October 2018.
201. "Skylark Drones set to raise its first round of funding to boost expansion" (<http://www.financialexpress.com/industry/companies/sounds-like-a-drone/135174/>). 14 September 2015. Archived (<https://web.archive.org/web/20160914170336/http://www.financialexpress.com/industry/companies/sounds-like-a-drone/135174/>) from the original on 14 September 2016. Retrieved 28 August 2016.

202. Peterson, Andrea (19 August 2013). "States are competing to be the Silicon Valley of drones" (<https://www.washingtonpost.com/news/the-switch/wp/2013/08/19/states-are-competing-to-be-the-silicon-valley-of-drones/>). *The Washington Post*. Archived (<https://web.archive.org/web/20160213121946/https://www.washingtonpost.com/news/the-switch/wp/2013/08/19/states-are-competing-to-be-the-silicon-valley-of-drones/>) from the original on 13 February 2016. Retrieved 4 February 2016.
203. "Drone Training Courses – The Complete List" (<http://dronebusinessmarketer.com/drone-training/>). *Drone Business Marketer*. Archived (<https://web.archive.org/web/20161116112235/http://dronebusinessmarketer.com/drone-training/>) from the original on 16 November 2016. Retrieved 1 December 2016.
204. "IDF buying mass-market DJI drones" (<https://web.archive.org/web/20171211053504/http://www.janes.com/article/71456/idf-buying-mass-market-dji-drones>). *Jane's 360*. Archived from the original (<http://www.janes.com/article/71456/idf-buying-mass-market-dji-drones>) on 11 December 2017.
205. Greenwood, Faine (16 August 2017). "The U.S. Military Shouldn't Use Commercial Drones" (https://web.archive.org/web/20180417154848/http://www.slate.com/articles/technology/future_tense/2017/08/the_u_s_military_shouldn_t_use_commercial_drones.html). *Slate*. Archived from the original (<https://slate.com/technology/2017/08/the-u-s-military-shouldnt-use-e-commercial-drones.html>) on 17 April 2018. Retrieved 2 June 2023.
206. "DJI Won the Drone Wars, and Now It's Paying the Price" (<https://www.bloomberg.com/news/features/2020-03-26/dji-s-drone-supremacy-comes-at-a-price>). *Bloomberg*. 26 March 2020. Archived (<https://web.archive.org/web/20201119092847/https://www.bloomberg.com/news/features/2020-03-26/dji-s-drone-supremacy-comes-at-a-price>) from the original on 19 November 2020. Retrieved 18 November 2020.
207. "大疆创新与新疆维吾尔自治区公安厅结为警用无人机战略合作伙伴" (<https://www.youuav.com/news/detail/201712/19315.html>). *YouUAV.com*. 24 December 2017. Archived (<https://web.archive.org/web/20201218075733/https://www.youuav.com/news/detail/201712/19315.html>) from the original on 18 December 2020. Retrieved 18 November 2020.
208. "The Next Frontier of Police Surveillance Is Drones" (<https://slate.com/technology/2018/06/axon-and-dji-are-teaming-up-to-make-surveillance-drones-and-the-possibilities-are-frightening.html>). *Slate*. 7 June 2018. Archived (<https://web.archive.org/web/20191211090916/https://slate.com/technology/2018/06/axon-and-dji-are-teaming-up-to-make-surveillance-drones-and-the-possibilities-are-frightening.html>) from the original on 11 December 2019. Retrieved 10 December 2019.
209. "These Police Drones are Watching You" (<https://www.pogo.org/analysis/2018/09/these-police-drones-are-watching-you/>). *Project On Government Oversight*. 25 September 2018. Archived (<https://web.archive.org/web/20191211163002/https://www.pogo.org/analysis/2018/09/these-police-drones-are-watching-you/>) from the original on 11 December 2019. Retrieved 10 December 2019.
210. Stockman, Farah (13 July 2025). "Drones Are Key to Winning Wars Now. The U.S. Makes Hardly Any" (<https://www.nytimes.com/2025/07/13/business/drones-us-military-manufacturing-lags.html>). *The New York Times*.
211. "AERIGON cinema drone (UAV) pioneering in film production" (<https://newtonnordic.com/aerigon-cinema-drone/>). Archived (<https://web.archive.org/web/20210826200035/https://newtonnordic.com/aerigon-cinema-drone/>) from the original on 26 August 2021. Retrieved 26 August 2021.
212. "Global Agriculture Drones and Robots Market Analysis & Forecast, 2018-2028 - ResearchAndMarkets.com" (<https://finance.yahoo.com/news/global-agriculture-drones-robots-market-101300454.html>). *finance.yahoo.com*. Archived (<https://web.archive.org/web/20190707150807/https://finance.yahoo.com/news/global-agriculture-drones-robots-market-101300454.html>) from the original on 7 July 2019. Retrieved 23 May 2019.

213. "Africa Farming Problems Aided With Drone Technology" (<https://web.archive.org/web/20180629235733/https://www.droneaddicts.net/africa-farming-problems-aided-with-drone-technology/>). *Drone Addicts*. 12 March 2018. Archived from the original (<https://www.droneaddicts.net/africa-farming-problems-aided-with-drone-technology/>) on 29 June 2018. Retrieved 23 May 2019.
214. Brocious, Ariana (3 August 2016). "Drones That Launch Flaming Balls Are Being Tested To Help Fight Wildfires" (<https://www.npr.org/sections/alltechconsidered/2016/08/03/488477317/drones-that-launch-flaming-balls-are-being-tested-to-help-fight-wildfires>). NPR. Archived (<https://web.archive.org/web/20180425164411/https://www.npr.org/sections/alltechconsidered/2016/08/03/488477317/drones-that-launch-flaming-balls-are-being-tested-to-help-fight-wildfires>) from the original on 25 April 2018. Retrieved 5 April 2018.
215. Old, Julie M.; Lin, Simon H.; Franklin, Michael J.M. (December 2019). "Mapping out bare-nosed wombat (*Vombatus ursinus*) burrows with the use of a drone" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6749681>). *BMC Ecology*. **19** (1) 39. Bibcode:2019BMCE...19...39O (<https://ui.adsabs.harvard.edu/abs/2019BMCE...19...39O>). doi:10.1186/s12898-019-0257-5 (<https://doi.org/10.1186/s12898-019-0257-5>). PMC 6749681 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6749681>). PMID 31533684 (<https://pubmed.ncbi.nlm.nih.gov/31533684>).
216. "Drone Light Shows Powered by Intel" (<https://www.intel.com/content/www/us/en/technology-innovation/aerial-technology-light-show.html>). *Intel*. Archived (<https://web.archive.org/web/20210623231138/https://www.intel.com/content/www/us/en/technology-innovation/aerial-technology-light-show.html>) from the original on 23 June 2021. Retrieved 28 June 2021.
217. Hirsch, Lauren (1 July 2023). "Fireworks Have a New Competitor: Drones" (<https://www.nytimes.com/2023/07/01/business/dealbook/fourth-of-july-fireworks-drones.html>). *The New York Times*. Retrieved 10 August 2023.
218. "Fireworks and Drones Combine to Create Amazing Long Exposure Images" (<https://mossandfog.com/fireworks-and-drones-combine-to-create-amazing-long-exposure-images/>). *Moss and Fog*. 1 May 2023. Retrieved 10 August 2023.
219. Eltner, Anette; Hoffmeister, Dirk; Kaiser, Andreas; Karrasch, Pierre; Klingbeil, Lasse; Stöcker, Claudia; Rovere, Alessio, eds. (2022). *UAVs for the environmental sciences: methods and applications*. Darmstadt: wbg Academic. ISBN 978-3-534-40588-6.
220. Ferreira, Edgar; Chandler, Jim; Wackrow, Rene; Shiono, Koji (April 2017). "Automated extraction of free surface topography using SfM-MVS photogrammetry" (<https://doi.org/10.1016%2Fj.flowmeasinst.2017.02.001>). *Flow Measurement and Instrumentation*. **54**: 243–249. Bibcode:2017FloMI..54..243F (<https://ui.adsabs.harvard.edu/abs/2017FloMI..54..243F>). doi:10.1016/j.flowmeasinst.2017.02.001 (<https://doi.org/10.1016%2Fj.flowmeasinst.2017.02.001>).
221. Reddy, C. Sudhakar; Kurian, Ayushi; Srivastava, Gaurav; Singhal, Jayant; Varghese, A. O.; Padalia, Hitendra; Ayyappan, N.; Rajashekar, G.; Jha, C. S.; Rao, P. V. N. (January 2021). "Remote sensing enabled essential biodiversity variables for biodiversity assessment and monitoring: technological advancement and potentials". *Biodiversity and Conservation*. **30** (1): 1–14. Bibcode:2021BiCon..30....1R (<https://ui.adsabs.harvard.edu/abs/2021BiCon..30....1R>). doi:10.1007/s10531-020-02073-8 (<https://doi.org/10.1007/s10531-020-02073-8>).
222. Gonçalves, João; Henriques, Renato; Alves, Paulo; Sousa-Silva, Rita; Monteiro, António T.; Lomba, Ângela; Marcos, Bruno; Honrado, João (January 2016). "Evaluating an unmanned aerial vehicle-based approach for assessing habitat extent and condition in fine-scale early successional mountain mosaics". *Applied Vegetation Science*. **19** (1): 132–146. Bibcode:2016AppVS..19..132G (<https://ui.adsabs.harvard.edu/abs/2016AppVS..19..132G>). doi:10.1111/avsc.12204 (<https://doi.org/10.1111%2Favsc.12204>).

223. Barbizan Sühs, Rafael; Ziller, Silvia R.; Dechoum, Michele (February 2024). "Is the use of drones cost-effective and efficient in detecting invasive alien trees? A case study from a subtropical coastal ecosystem". *Biological Invasions*. **26** (2): 357–363. Bibcode:2024BiInv..26..357B (<https://ui.adsabs.harvard.edu/abs/2024BiInv..26..357B>). doi:10.1007/s10530-023-03190-5 (<https://doi.org/10.1007%2Fs10530-023-03190-5>).
224. Zhang, Chunhua; Kovacs, John M. (December 2012). "The application of small unmanned aerial systems for precision agriculture: a review". *Precision Agriculture*. **13** (6): 693–712. Bibcode:2012PrAgr..13..693Z (<https://ui.adsabs.harvard.edu/abs/2012PrAgr..13..693Z>). doi:10.1007/s11119-012-9274-5 (<https://doi.org/10.1007%2Fs11119-012-9274-5>).
225. Perks, Matthew T.; Russell, Andrew J.; Large, Andrew R. G. (5 October 2016). "Technical Note: Advances in flash flood monitoring using unmanned aerial vehicles (UAVs)" (<https://doi.org/10.5194%2Fhess-20-4005-2016>). *Hydrology and Earth System Sciences*. **20** (10): 4005–4015. Bibcode:2016HESS...20.4005P (<https://ui.adsabs.harvard.edu/abs/2016HES S...20.4005P>). doi:10.5194/hess-20-4005-2016 (<https://doi.org/10.5194%2Fhess-20-4005-2016>).
226. Zhou, Jianguo; He, Linshu; Luo, Haitao (19 March 2023). "Real-Time Positioning Method for UAVs in Complex Structural Health Monitoring Scenarios" (<https://doi.org/10.3390%2Fdrone s7030212>). *Drones*. **7** (3): 212. Bibcode:2023Drone...7..212Z (<https://ui.adsabs.harvard.edu/abs/2023Drone...7..212Z>). doi:10.3390/drones7030212 (<https://doi.org/10.3390%2Fdrone s7030212>).
227. Jackisch, Robert; Lorenz, Sandra; Zimmermann, Robert; Möckel, Robert; Gloaguen, Richard (2018). "Drone-Borne Hyperspectral Monitoring of Acid Mine Drainage: An Example from the Sokolov Lignite District" (<https://doi.org/10.3390%2Frs10030385>). *Remote Sensing*. **10** (3): 385. Bibcode:2018RemS...10..385J (<https://ui.adsabs.harvard.edu/abs/2018RemS...10..385J>). doi:10.3390/rs10030385 (<https://doi.org/10.3390%2Frs10030385>).
228. Flores, Hernan; Lorenz, Sandra; Jackisch, Robert; Tusa, Laura; Contreras, I. Cecilia; Zimmermann, Robert; Gloaguen, Richard (2021). "UAS-Based Hyperspectral Environmental Monitoring of Acid Mine Drainage Affected Waters" (<https://doi.org/10.3390%2Fmin11020182>). *Minerals*. **11** (2): 182. Bibcode:2021Mine...11..182F (<https://ui.adsabs.harvard.edu/abs/2021Mine...11..182F>). doi:10.3390/min11020182 (<https://doi.org/10.3390%2Fmin11020182>).
229. Karbach, Niklas (2022). "Observing volcanoes with drones: studies of volcanic plume chemistry with ultralight sensor systems" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9596470>). *Scientific Reports*. **12** (17890) 17890. Bibcode:2022NatSR..1217890K (<https://ui.adsabs.harvard.edu/abs/2022NatSR..1217890K>). doi:10.1038/s41598-022-21935-5 (<https://doi.org/10.1038%2Fs41598-022-21935-5>). PMC 9596470 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9596470>). PMID 36284218 (<https://pubmed.ncbi.nlm.nih.gov/36284218>).
230. Sun, Jianwei; Yuan, Guoqin; Song, Laiyun; Zhang, Hongwen (January 2024). "Unmanned Aerial Vehicles (UAVs) in Landslide Investigation and Monitoring: A Review" (<https://doi.org/10.3390%2Fdrone s8010030>). *Drones*. **8** (1): 30. Bibcode:2024Drone...8...30S (<https://ui.adsabs.harvard.edu/abs/2024Drone...8...30S>). doi:10.3390/drones8010030 (<https://doi.org/10.3390%2Fdrone s8010030>).
231. Dai, Keren; Li, Zhiyu; Xu, Qiang; Tomas, Roberto; Li, Tao; Jiang, Liming; Zhang, Jianyong; Yin, Tao; Wang, Hao (July 2023). "Identification and evaluation of the high mountain upper slope potential landslide based on multi-source remote sensing: the Aniangzhai landslide case study". *Landslides*. **20** (7): 1405–1417. Bibcode:2023Lands..20.1405D (<https://ui.adsabs.harvard.edu/abs/2023Lands..20.1405D>). doi:10.1007/s10346-023-02044-4 (<https://doi.org/10.1007%2Fs10346-023-02044-4>). hdl:10045/133124 (<https://hdl.handle.net/10045%2F133124>).

232. Yang, Yuchuan; Wang, Xiaobo; Jin, Wei; Cao, Jiayun; Cheng, Baogen; Zhou, Shunwen (October 2019). "Characteristics analysis of the reservoir landslides base on unmanned aerial vehicle (UAV) scanning technology at the Maoergai Hydropower Station, Southwest China" (<https://doi.org/10.1088%2F1755-1315%2F349%2F1%2F012009>). *IOP Conference Series: Earth and Environmental Science*. **349** (1) 012009. Bibcode:2019E&ES..349a2009Y (<https://ui.adsabs.harvard.edu/abs/2019E&ES..349a2009Y>). doi:10.1088/1755-1315/349/1/012009 (<https://doi.org/10.1088%2F1755-1315%2F349%2F1%2F012009>).
233. Tomás, Roberto; Pinheiro, Marisa; Pinto, Pedro; Pereira, Eduardo; Miranda, Tiago (August 2023). "Preliminary analysis of the mechanisms, characteristics, and causes of a recent catastrophic structurally controlled rock planar slide in Esposende (northern Portugal)" (<https://doi.org/10.1007%2Fs10346-023-02082-y>). *Landslides*. **20** (8): 1657–1665. Bibcode:2023Lands..20.1657T (<https://ui.adsabs.harvard.edu/abs/2023Lands..20.1657T>). doi:10.1007/s10346-023-02082-y (<https://doi.org/10.1007%2Fs10346-023-02082-y>). hdl:1822/88576 (<https://hdl.handle.net/1822%2F88576>).
234. Zhou, Jiawen; Jiang, Nan; Li, Congjiang; Li, Haibo (October 2024). "A landslide monitoring method using data from unmanned aerial vehicle and terrestrial laser scanning with insufficient and inaccurate ground control points" (<https://doi.org/10.1016%2Fj.jrmge.2023.12.004>). *Journal of Rock Mechanics and Geotechnical Engineering*. **16** (10): 4125–4140. Bibcode:2024JRMGE..16.4125Z (<https://ui.adsabs.harvard.edu/abs/2024JRMGE..16.4125Z>). doi:10.1016/j.jrmge.2023.12.004 (<https://doi.org/10.1016%2Fj.jrmge.2023.12.004>).
235. Peterman, V. (26 August 2015). "Landslide Activity Monitoring with the Help of Unmanned Aerial Vehicle" (<https://doi.org/10.5194%2Fisprsarchives-XL-1-W4-215-2015>). *The International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences*. XL-1/W4: 215–218. Bibcode:2015ISPAr.XL1..215P (<https://ui.adsabs.harvard.edu/abs/2015ISPAr.XL1..215P>). doi:10.5194/isprsarchives-XL-1-W4-215-2015 (<https://doi.org/10.5194%2Fisprsarchives-XL-1-W4-215-2015>).
236. Sindi & Zarei (15 September 2023). "Drones in humanitarian aid – Can they be a game-changer?" (<https://web.archive.org/web/20231116094456/https://blogs.coventry.ac.uk/researchblog/drones-in-humanitarian-aid-can-they-be-a-game-changer/>). CURB. Archived from the original (<https://blogs.coventry.ac.uk/researchblog/drones-in-humanitarian-aid-can-they-be-a-game-changer/>) on 16 November 2023. Retrieved 16 November 2023.
237. Faust, Daniel R. (2015). *Police Drones*. The Rosen Publishing Group, Inc. ISBN 978-1-5081-4502-8.
238. Heincke, Björn; Jackisch, Robert; Saartenoja, Ari; Salmirinne, Heikki; Rapp, Sönke; Zimmermann, Robert; Pirttijärvi, Markku; Vest Sörensen, Erik; Gloaguen, Richard; Ek, Lisa; Bergström, Johan; Karinen, Arto; Salehi, Sara; Madriz, Yuleika; Middleton, Maarit (29 July 2019). "Developing multi-sensor drones for geological mapping and mineral exploration: setup and first results from the MULSEDRO project". *Geological Survey of Denmark and Greenland Bulletin*. **43**. doi:10.34194/GEUSB-201943-03-02 (<https://doi.org/10.34194%2FGEUSB-201943-03-02>).
239. Šálek, Ondřej; Matolín, Milan; Gryc, Lubomír (February 2018). "Mapping of radiation anomalies using UAV mini-airborne gamma-ray spectrometry". *Journal of Environmental Radioactivity*. **182**: 101–107. Bibcode:2018JEnvR.182..101S (<https://ui.adsabs.harvard.edu/abs/2018JEnvR.182..101S>). doi:10.1016/j.jenvrad.2017.11.033 (<https://doi.org/10.1016%2Fj.jenvrad.2017.11.033>). PMID 29220714 (<https://pubmed.ncbi.nlm.nih.gov/29220714>).
240. Jackisch, Robert (17 March 2020). "Drone-based surveys of mineral deposits". *Nature Reviews Earth & Environment*. **1** (4): 187. Bibcode:2020NRvEE...1..187J (<https://ui.adsabs.harvard.edu/abs/2020NRvEE...1..187J>). doi:10.1038/s43017-020-0042-1 (<https://doi.org/10.1038%2Fs43017-020-0042-1>).

241. Jackisch, Robert; Heincke, Björn H.; Zimmermann, Robert; Sørensen, Erik V.; Pirttijärvi, Markku; Kirsch, Moritz; Salmirinne, Heikki; Lode, Stefanie; Kuronen, Urpo; Gloaguen, Richard (7 April 2022). "Drone-based magnetic and multispectral surveys to develop a 3D model for mineral exploration at Qullissat, Disko Island, Greenland" (<https://doi.org/10.5194/se-13-793-2022>). *Solid Earth*. **13** (4): 793–825. Bibcode:2022SolE...13..793J (<https://ui.adsabs.harvard.edu/abs/2022SolE...13..793J>). doi:10.5194/se-13-793-2022 (<https://doi.org/10.5194/se-13-793-2022>). hdl:11250/3044853 (<https://hdl.handle.net/11250/3044853>).
242. Dent, Steve (16 October 2017). "Drone hits a commercial plane for the first time in Canada" (<https://web.archive.org/web/20171016170044/https://finance.yahoo.com/news/drone-hits-commercial-plane-first-134300420.html>). Engadget. Archived from the original (<https://finance.yahoo.com/news/drone-hits-commercial-plane-first-134300420.html>) on 16 October 2017. Retrieved 16 October 2017.
243. Tellman, Julie (28 September 2018). "First-ever recorded drone-hot air balloon collision prompts safety conversation" (https://www.postregister.com/news/local/first-ever-recorded-drone-hot-air-balloon-collision-prompts-safety/article_7cc41c24-6025-5aa6-b6dd-6d1ea5e85961.html). *Teton Valley News*. Boise, Idaho, United States: Boise Post-Register. Archived (https://web.archive.org/web/20181003220947/https://www.postregister.com/news/local/first-ever-recorded-drone-hot-air-balloon-collision-prompts-safety/article_7cc41c24-6025-5aa6-b6dd-6d1ea5e85961.html) from the original on 3 October 2018. Retrieved 3 October 2018.
244. "Drones need to be encouraged, and people protected" (<https://www.economist.com/leaders/2019/01/26/drones-need-to-be-encouraged-and-people-protected>). *The Economist*. 26 January 2019. ProQuest 2171135630 (<https://www.proquest.com/docview/2171135630>). Archived (<https://web.archive.org/web/20200627180313/https://www.economist.com/leaders/2019/01/26/drones-need-to-be-encouraged-and-people-protected>) from the original on 27 June 2020. Retrieved 28 June 2020.
245. Halon, Eytan (21 December 2018). "Israeli anti-drone technology brings an end to Gatwick Airport chaos – International news – Jerusalem Post" (<https://www.jpost.com/International/Israeli-anti-drone-technology-brings-an-end-to-Gatwick-Airport-chaos-575054>). *jpost.com*. Archived (<https://web.archive.org/web/20181222131417/https://www.jpost.com/International/Israeli-anti-drone-technology-brings-an-end-to-Gatwick-Airport-chaos-575054>) from the original on 22 December 2018. Retrieved 22 December 2018.
246. Matthew Weaver; Damien Gayle; Patrick Greenfield; Frances Perraudin (20 December 2018). "Military called in to help with Gatwick drone crisis" (<https://www.theguardian.com/uk-news/2018/dec/19/gatwick-flights-halted-after-drone-sighting>). *The Guardian*. Archived (<https://web.archive.org/web/20181222023943/https://www.theguardian.com/uk-news/2018/dec/19/gatwick-flights-halted-after-drone-sighting>) from the original on 22 December 2018. Retrieved 22 December 2018.
247. Graf, Scott (24 July 2015). "In The Heat of the Moment, Drones Are Getting in the Way of Firefighters" (<https://www.npr.org/sections/alltechconsidered/2015/07/24/425652212/in-the-heat-of-the-moment-drones-are-getting-in-the-way-of-firefighters>). NPR. Archived (<https://web.archive.org/web/20180305155859/https://www.npr.org/sections/alltechconsidered/2015/07/24/425652212/in-the-heat-of-the-moment-drones-are-getting-in-the-way-of-firefighters>) from the original on 5 March 2018. Retrieved 5 April 2018.
248. Michael Martinez; Paul Vercammen; Ben Brumfield (18 July 2015). "Drones visit California wildfire, angering firefighters" (<http://www.cnn.com/2015/07/18/us/california-freeway-fire/index.html>). CNN. Archived (<https://web.archive.org/web/20161108102742/http://www.cnn.com/2015/07/18/us/california-freeway-fire/index.html>) from the original on 8 November 2016. Retrieved 22 August 2016.

249. Medina, Jennifer (19 July 2015). "Chasing Video With Drones, Hobbyists Imperil California Firefighting Efforts" (<https://www.nytimes.com/2015/07/20/us/hobby-drones-hinder-california-firefighting-efforts.html>). *The New York Times*. Archived (<https://web.archive.org/web/20150721041118/http://www.nytimes.com/2015/07/20/us/hobby-drones-hinder-california-firefighting-efforts.html>) from the original on 21 July 2015 – via NYTimes.com.
250. Rocha, Veronica (21 July 2015). "Attack on the drones: Legislation could allow California firefighters to take them down" (<http://www.latimes.com/local/lanow/la-me-ln-anti-drone-legislation-20150721-story.html>). Archived (<https://web.archive.org/web/20160828153646/http://www.latimes.com/local/lanow/la-me-ln-anti-drone-legislation-20150721-story.html>) from the original on 28 August 2016. Retrieved 22 August 2016 – via LA Times.
251. Samilton, Tracy (15 November 2017). "Prisons Work To Keep Out Drug-Smuggling Drones" (<https://www.npr.org/2017/11/15/564272346/prisons-work-to-keep-out-drug-smuggling-drones>). NPR. Archived (<https://web.archive.org/web/20180119175506/https://www.npr.org/2017/11/15/564272346/prisons-work-to-keep-out-drug-smuggling-drones>) from the original on 19 January 2018. Retrieved 19 January 2018.
252. Mike Mount; Elaine Quijano. "Iraqi insurgents hacked Predator drone feeds, U.S. official indicates" (<http://edition.cnn.com/2009/US/12/17/drone.video.hacked/>). *CNN.com*. Archived (<https://web.archive.org/web/20170305082148/http://edition.cnn.com/2009/US/12/17/drone.video.hacked/>) from the original on 5 March 2017. Retrieved 6 December 2016.
253. Walters, Sander (29 October 2016). "How Can Drones Be Hacked? The updated list of vulnerable drones & attack tools" (<https://medium.com/@swalters/how-can-drones-be-hacked-the-updated-list-of-vulnerable-drones-attack-tools-dd2e006d6809>). *Medium*. Archived (<https://web.archive.org/web/20180723003615/https://medium.com/@swalters/how-can-drones-be-hacked-the-updated-list-of-vulnerable-drones-attack-tools-dd2e006d6809>) from the original on 23 July 2018. Retrieved 6 December 2016.
254. Glaser, April (4 January 2017). "The U.S. government showed just how easy it is to hack drones made by Parrot, DBPower and Cheerson" (<http://www.recode.net/2017/1/4/14062654/drones-hacking-security-ftc-parrot-dbpower-cheerson>). *Recode*. Archived (<https://web.archive.org/web/20170105190345/http://www.recode.net/2017/1/4/14062654/drones-hacking-security-ftc-parrot-dbpower-cheerson>) from the original on 5 January 2017. Retrieved 6 January 2017.
255. Rebecca L. Scharf, Drone Invasion: Unmanned Aerial Vehicles and the Right to Privacy (<https://scholars.law.unlv.edu/cgi/viewcontent.cgi?article=2240&context=facpub>), INDIANA LAW JOURNAL Vol. 94, 2019
256. "Anti-drone technology to be test flown on UK base amid terror fears" (<https://eandt.theiet.org/content/articles/2017/03/anti-drone-technology-to-be-test-flown-on-uk-base-amid-terror-fears/>). 6 March 2017. Archived (<https://web.archive.org/web/20170507230433/https://eandt.theiet.org/content/articles/2017/03/anti-drone-technology-to-be-test-flown-on-uk-base-amid-terror-fears/>) from the original on 7 May 2017. Retrieved 9 May 2017.
257. Isaac-Medina, Brian K. S.; Poyser, Matthew; Organisciak, Daniel; Willcocks, Chris G.; Breckon, Toby P.; Shum, Hubert P. H. (2021). *Unmanned Aerial Vehicle Visual Detection and Tracking using Deep Neural Networks: A Performance Benchmark*. pp. 1223–1232. arXiv:2103.13933 (<https://arxiv.org/abs/2103.13933>).
258. Organisciak, Daniel; Poyser, Matthew; Alsehaim, Aishah; Hu, Shanfeng; Isaac-Medina, Brian K. S.; Breckon, Toby P.; Shum, Hubert P. H. (2022). "UAV-ReID: A Benchmark on Unmanned Aerial Vehicle Re-identification in Video Imagery". *Proceedings of the 17th International Joint Conference on Computer Vision, Imaging and Computer Graphics Theory and Applications*. SciTePress. pp. 136–146. arXiv:2104.06219 (<https://arxiv.org/abs/2104.06219>). doi:10.5220/0010836600003124 (<https://doi.org/10.5220/0010836600003124>). ISBN 978-989-758-555-5.

259. "Heathrow picks C-UAS to combat drone disruption" (<https://www.shephardmedia.com/news/uv-online/heathrows-picks-new-c-uas-combat-drone-disruption/>). Archived (<https://web.archive.org/web/20191109111036/https://www.shephardmedia.com/news/uv-online/heathrows-picks-new-c-uas-combat-drone-disruption/>) from the original on 9 November 2019. Retrieved 13 March 2019.
260. "Muscat International Airport to install USD10 million Aaronia counter-UAS system" (<https://www.unmannedairspace.info/counter-uas-systems-and-policies/muscat-international-airport-install-usd10-million-aaronia-counter-uas-system/>). 21 January 2019. Archived (<https://web.archive.org/web/20191109111704/https://www.unmannedairspace.info/counter-uas-systems-and-policies/muscat-international-airport-install-usd10-million-aaronia-counter-uas-system/>) from the original on 9 November 2019. Retrieved 21 January 2019.
261. Grand-Clément, Sarah; Bajon, Theò (19 October 2022). "Uncrewed Aerial Systems: A Primer" (<https://web.archive.org/web/20230105181006/https://unidir.org/publication/uncrewed-aerial-systems-primer>). *United Nations Institute for Disarmament Research*. Archived from the original (<https://unidir.org/publication/uncrewed-aerial-systems-primer>) on 5 January 2023. Retrieved 5 January 2023.
262. Hartley, John; Shum, Hubert P.H.; Ho, Edmond S.L.; Wang, He; Ramamoorthy, Subramanian (2022). "Formation Control for UAVs Using a Flux Guided Approach". *Expert Systems with Applications*. **205** 117665. arXiv:2103.09184 (<https://arxiv.org/abs/2103.09184>). doi:10.1016/j.eswa.2022.117665 (<https://doi.org/10.1016%2Fj.eswa.2022.117665>).
263. "Anti-Drone Protection" (<https://www.elbitsystems.com/land/land-ew-sigint/anti-drone-protection>). *Elbit Systems*. Retrieved 5 June 2025.
264. "ReDrone Portable CUAS" (<https://www.elbitsystems.com/homeland-security/observation-surveillance-detection/anti-drone-systems/redrone-portable-cuas>). *Elbit Systems*. Retrieved 5 June 2025.
265. "Red Sky 2" (<https://www.elbitsystems.com/homeland-security/observation-surveillance-detection/anti-drone-systems>). *Elbit Systems*. Retrieved 5 June 2025.
266. "What is unmanned traffic management?" (<https://www.airbus.com/newsroom/stories/what-is-unmanned-traffic-management-utm.html>). *Airbus*. Archived (<https://web.archive.org/web/20210210208195828/https://www.airbus.com/newsroom/stories/what-is-unmanned-traffic-management-utm.html>) from the original on 8 February 2021. Retrieved 28 January 2021.
267. Cary, Leslie; Coyne, James. "ICAO Unmanned Aircraft Systems (UAS), Circular 328". *2011–2012 UAS Yearbook – UAS: The Global Perspective* (https://web.archive.org/web/20160304025432/http://uvs-info.com/phocadownload/05_3a_2011/P112-P115_C%26AI_ICAO-Advisory-Circular.pdf) (PDF). Blyenburgh & Co. pp. 112–115. Archived from the original (http://uvs-info.com/phocadownload/05_3a_2011/P112-P115_C&AI_ICAO-Advisory-Circular.pdf) (PDF) on 4 March 2016. Retrieved 26 February 2022.
268. Boedecker, Hendrik. "The 2021 Drone Regulation – What is new? What is planned?" (<https://droneii.com/the-2021-drone-regulation-what-is-new-what-is-planned>). *Drone Industry Insights*. Archived (<https://web.archive.org/web/20210517121444/https://droneii.com/the-2021-drone-regulation-what-is-new-what-is-planned>) from the original on 17 May 2021. Retrieved 17 May 2021.
269. "UAS Remote Identification Overview" (https://www.faa.gov/uas/getting_started/remote_id/). *www.faa.gov*. Archived (https://web.archive.org/web/20210527012153/https://www.faa.gov/uas/getting_started/remote_id/) from the original on 27 May 2021. Retrieved 29 May 2021.
270. "FAA Legal Battle – Challenging Remote ID" (<https://www.racedayquads.com/pages/faa-legal-battle-to-save-fpv>). *RaceDayQuads*. Archived (<https://web.archive.org/web/20210527012139/https://www.racedayquads.com/pages/faa-legal-battle-to-save-fpv>) from the original on 27 May 2021. Retrieved 29 May 2021.

271. Dong, Joy (5 April 2026). "China Built the World's Drone Industry. Now It's Locking Down the Skies" (<https://www.nytimes.com/2026/04/05/world/asia/china-drone-regulations.html>). *The New York Times*. ISSN 0362-4331 (<https://search.worldcat.org/issn/0362-4331>). Retrieved 6 April 2026.
272. "UAS Class Label" (<https://eudroneport.com/blog/uas-class-label/>). *www.eudroneport.com*. August 2022. Archived (<https://web.archive.org/web/20221005144734/https://eudroneport.com/blog/uas-class-label/>) from the original on 5 October 2022. Retrieved 21 February 2023.
273. Consolidated text: Commission Delegated Regulation (EU) 2019/945 of 12 March 2019 on unmanned aircraft systems and on third-country operators of unmanned aircraft systems (<https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:02019R0945-20200809>)

Sources

- Sayler, Kelley (June 2015). "A world of proliferated drones: a technology primer" (https://web.archive.org/web/20160306050607/http://www.cnas.org/sites/default/files/publications-pdf/CNAS%20World%20of%20Drones_052115.pdf) (PDF). *Center for a New American Security*. Archived from the original (http://www.cnas.org/sites/default/files/publications-pdf/CNAS%20World%20of%20Drones_052115.pdf) (PDF) on 6 March 2016.
- Wagner, William (1982). *Lightning Bugs and Other Reconnaissance Drones*. Washington, D.C.: Armed Forces Journal International, in cooperation with Aero Publishers. ISBN 978-0-8168-6654-0.

Further reading

- Axe, David (2021). *Drone War Vietnam*. Yorkshire: Pen & Sword Military. ISBN 978-1-5267-7026-4.
- Cahill, Bill (April 2022). "Lightning Bugs & Buffalo Hunters: The Ryan Model 147 Drone in Vietnam". *The Aviation Historian* (39): 18–27.
- El-Sayed, Ahmed F. (2025). "History and Classification of UAV". *History and Evolution of Aircraft: Technological Advancements in Size, Speed, Armaments, and Engines* (1st ed.). Boca Raton, FL: CRC Press. pp. 306–346. doi:10.1201/9781003450115-8 (<https://doi.org/10.1201/9781003450115-8>). ISBN 9781003450115.
- Hill, John; Rogers, Ann (2014). *The rise of the drones: From The Great War to Gaza*. Arts & Humanities Colloquium Series. Vancouver Island University. hdl:10613/2480 (<https://hdl.handle.net/10613/2480>).
- Garcia-Bernardo, Javier; Dodds, Peter Sheridan; Johnson, Neil F. (February 2016). "Quantitative patterns in drone wars". *Physica A: Statistical Mechanics and Its Applications*. **443**: 380–384. Bibcode:2016PhyA..443..380G (<https://ui.adsabs.harvard.edu/abs/2016PhyA..443..380G>). doi:10.1016/j.physa.2015.09.055 (<https://doi.org/10.1016/j.physa.2015.09.055>).
- Mohsan, Syed Agha Hassnain; Khan, Muhammad Asghar; Noor, Fazal; Ullah, Insaf; Alsharif, Mohammed H. (2022). "Towards the Unmanned Aerial Vehicles (UAVs): A Comprehensive Review" (<https://doi.org/10.3390/drones6060147>). *Drones*. **6** (6) 147. Bibcode:2022Drone...6..147M (<https://ui.adsabs.harvard.edu/abs/2022Drone...6..147M>). doi:10.3390/drones6060147 (<https://doi.org/10.3390/drones6060147>).
- Rogers, Ann; Hill, John (2014). *Unmanned: Drone warfare and global security*. Between the Lines. ISBN 978-1-77113-154-4.
- Singh, Ruchita; Kumar, Sandeep (2025). "A Comprehensive Insights into Drones: History, Classification, Architecture, Navigation, Applications, Challenges, and Future Trends".

arXiv:2501.10066 (<https://arxiv.org/abs/2501.10066>) [cs.RO (<https://arxiv.org/archive/cs.RO>)].

- Mohsan, Syed Agha Hassnain; Othman, Nawaf Qasem Hamood; Li, Yanlong; Alsharif, Mohammed H.; Khan, Muhammad Asghar (2023). "Unmanned aerial vehicles (UAVs): practical aspects, applications, open challenges, security issues, and future trends" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9841964>). *Intelligent Service Robotics*. **16** (1): 109–137. doi:10.1007/s11370-022-00452-4 (<https://doi.org/10.1007/s11370-022-00452-4>). PMC 9841964 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9841964>). PMID 36687780 (<https://pubmed.ncbi.nlm.nih.gov/36687780>).

External links

- How Intelligent Drones Are Shaping the Future of Warfare (<https://www.rollingstone.com/culture/features/how-intelligent-drones-are-shaping-the-future-of-warfare-w471703>) Archived (<https://web.archive.org/web/20180502113749/https://www.rollingstone.com/culture/features/how-intelligent-drones-are-shaping-the-future-of-warfare-w471703>) 2 May 2018 at the Wayback Machine, *Rolling Stone*

Retrieved from "https://en.wikipedia.org/w/index.php?title=Unmanned_aerial_vehicle&oldid=1377209233"